

Abundance in numerical dimension one for terminal threefolds in positive characteristic

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Abstract

Let X be a projective $\mathbb{Q}$ -factorial terminal threefold over an algebraically closed field of characteristic $p > 3$. We prove the numerical-dimension-one case of abundance: if K_X is nef with $\nu(K_X) = 1$, then K_X is semiample and $\kappa(X, K_X) = 1$.

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1 Introduction

The abundance conjecture predicts that a nef canonical divisor on a projective variety with mild singularities is semiample: some positive Cartier multiple is generated by its global sections. Such sections turn numerical information about the canonical divisor into a morphism that records the canonical geometry of the variety. For a nef divisor L on a threefold, numerical dimension one means $L \cdot H^2 > 0$ and $L^2 \cdot H = 0$ for an ample Cartier divisor H . We prove this case of abundance for terminal threefolds in characteristic greater than three.

Theorem 1.1. *Let X be a normal projective $\mathbb{Q}$ -factorial threefold over an algebraically closed field k of characteristic $p > 3$. Suppose that X has terminal singularities and that K_X is nef with $\nu(K_X) = 1$. Then some positive Cartier multiple of K_X has at least two independent global sections. In particular, $\kappa(X, K_X) = 1$ and K_X is semiample.*

No assumption is made on the first Betti number, irregularity, or Albanese variety. Thus Theorem 1.1 resolves the numerical-dimension-one case of the abundance conjecture under the stated terminal and characteristic hypotheses.

Geometrically, the theorem gives a morphism $f : X \rightarrow C$ with connected fibers to a normal projective curve and an ample Cartier divisor A on C such that $mK_X \sim f^*A$ for some $m > 0$. Indeed, the image of a morphism defined by a globally generated pluricanonical

multiple is a curve, since $\nu(K_X) = 1$. Its Stein factorization gives f ; the finite pullback to C of a hyperplane divisor on the image gives A .

1.1 Context

Over the complex numbers, Miyaoka proved the numerical-dimension-one case of threefold abundance by studying effective pluricanonical divisors, primitive connected cycles, and their infinitesimal neighborhoods [25]. Kawamata’s abundance theorem for minimal threefolds includes another proof of this case using the log minimal model program and formal thickenings [20, Section 4]. The present argument develops this canonical-cycle framework in positive characteristic. More recently, Liu–Xu developed higher-dimensional numerical-dimension-one abundance over $\mathbb{C}$ under nonvanishing hypotheses [24]. The ground field in the present theorem requires the positive-characteristic inputs described next.

The positive-characteristic minimal model program supplies the birational operations needed to study abundance for threefolds. Over algebraically closed fields of characteristic greater than five, Hacon–Xu established standard-coefficient flips and minimal models for projective $\mathbb{Q}$ -factorial terminal threefolds with pseudoeffective canonical divisor [12, Theorems 1.1 and 1.2]. Birkar proved flips for $\mathbb{Q}$ -factorial dlt threefold pairs and log minimal models for projective klt threefold pairs with pseudoeffective log canonical divisor [1, Theorems 1.1 and 1.2]. The relative log minimal model program in characteristic greater than five was established by Hashizume, Nakamura and Tanaka [16]; Hacon and Witaszek treated the characteristic-five case [11].

In characteristic greater than five, Xu–Zhang proved nonvanishing for terminal minimal threefolds [31]; Witaszek obtained nonvanishing for klt threefold pairs and abundance in nef dimension at most two through a canonical bundle formula [30]. Zhang proved abundance for minimal threefolds with nontrivial Albanese map [34]. Xu proved nonvanishing for pseudoeffective log canonical threefold pairs in characteristic greater than three and semiample when the nef log canonical divisor has positive Kodaira dimension [32, Theorems 1.4 and 1.6]. His theorem for nef dimension at most two [32, Theorem 1.8] and his subsequent numerical-dimension-two abundance theorem [33, Theorem 1.1] leave a specific issue in the present setting: an effective canonical divisor of numerical dimension one could a priori still have Kodaira dimension zero and maximal nef dimension. Theorem 1.1 excludes that possibility. Here nef dimension is the dimension of the nef reduction, and maximal nef dimension means that no curve of canonical degree zero passes through a very general point. It is a different invariant from numerical dimension. The new step is the production of a second pluricanonical section; the existence of a first section and the passage from positive Kodaira dimension to semiample are Xu’s results.

Our tangent-sheaf argument follows the positive-characteristic foliation method appearing in Xu–Zhang [31, Section 2]. It uses the Bogomolov inequality for strongly semistable sheaves with respect to a nef tuple, in Langer’s formulation [23, Theorem 3.2]; the corrections in [22] preserve that theorem. Surface abundance enters through abundance for projective semi-log-canonical surfaces, proved by Tanaka over algebraically closed fields [27, Theorem 0.1] and by Posva over arbitrary fields [26, Theorem 3.1].

The descent mechanism is classical Cartier theory. In particular, Katz’s proof of Cartier descent supplies the horizontal projector and truncated Taylor expansion [19, Theorem 5.1]. We establish the completed-ring and bounded-pole statements that allow this algebra to be used on the punctured formal neighborhood of a nonreduced cycle. The three-row jet calculation and its uniform pole control are the bridge between the canonical-cycle geometry and this descent argument.

1.2 The argument

After nonvanishing and the birational preparation in Section 2, the hypothetical maximal-nef-dimension case produces a smooth projective threefold V and a connected effective nef Cartier cycle D with primitive multiplicities. The cycle is numerically trivial on each of its components, its normal line $\mathcal{O}_D(D)$ is torsion, and, on a neighborhood of D , a p -power of ω_V has a divisor supported on D . The preparation proves torsion on the full nonreduced cycle, which is needed in the subsequent calculation.

Section 3 then proves strong semistability of the tangent and cotangent bundles for the nef degree $c_1(-) \cdot H \cdot D$. The foliation quotient of a destabilizing sheaf would produce D -trivial curves through very general points. The resulting semistability gives $c_2(V) \cdot D \geq 0$ and uniform bounds on polar parts, which are the inputs to the jet calculation.

The central calculation in Section 4 concerns finite jets along D , that is, sections on successive Cartier thickenings. It keeps all three cohomological rows of the pole filtration and compares their ranks over the periods rp^e , where r is the order of $\mathcal{O}_D(D)$. The scalar rows identify the completed canonical line with the line of an integral divisor supported on D , and force both $c_2(V) \cdot D = 0$ and the precise limiting ratio $p/(p+1)$ for the last trivialization widths. These facts control the positions and lengths of differentials for vector bundles as well.

Section 5 turns the calculation into a formal-neighborhood criterion, Theorem 5.2: a bundle satisfying an Euler identity, bounded polar parts, and a concentration condition for leading jets has vanishing first Laurent cohomology. A bound on the source cuts of incoming differentials ensures that the constructed primitives actually converge with bounded poles. We prove this criterion directly for finite covers and completed lattices. The criterion applies to one-forms and exact forms.

Section 6 uses these vanishings to make every line bundle on the punctured formal neighborhood a p th power, by Cartier descent. On the other hand, intersection with H on the components of D defines a nonzero integer-valued homomorphism on its Picard group. Such a homomorphism cannot exist on a p -divisible group. The contradiction proves Theorem 1.1.

Figure 1 records the two conclusions about the same Picard group that give the contradiction. In the figure, $\mathcal{R} = \hat{\mathcal{O}}_V(*D)$ denotes functions on the completion along D with a finite pole bound; its Picard group consists of locally free rank-one $\mathcal{R}$ -modules on $|D|$.

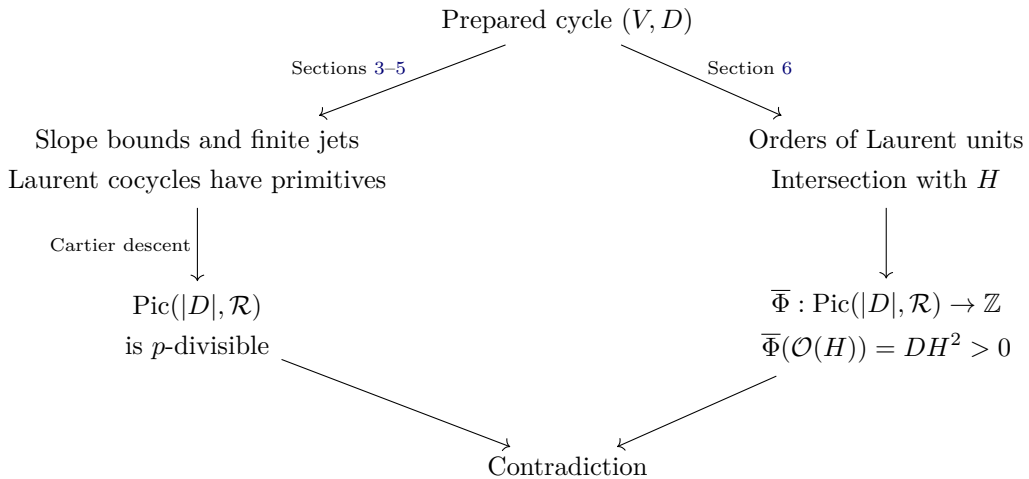


Figure 1: The maximal-nef-dimension assumption produces a cycle for which Laurent cohomology forces Picard divisibility, while intersection theory supplies a nonzero integral degree.

All arguments take place in characteristic p . The proof uses no lifting to characteristic

zero and no orbifold Iitaka theorem.

1.3 Conventions

Varieties are integral and all divisors are rational Cartier divisors when a pullback or intersection requires this, unless they are explicitly declared integral Cartier divisors. On a smooth variety we identify Cartier and Weil divisors. Linear and numerical equivalence are denoted by $\sim$ and $\equiv$, with a subscript $\mathbb{Q}$ for rational linear equivalence. For a nef divisor L , its nef dimension $n(L)$ is the dimension of its nef reduction; in particular, $n(L) = \dim X$ means that no L -trivial curve passes through a very general point. We use an uncountable algebraically closed field when making this formulation, and descend the final section-space statement to the original field.

If E is a vector bundle on a smooth threefold V , write $E^\dagger = E^\vee \otimes \omega_V$. All completions below are along an effective Cartier divisor, and all Laurent sheaves are considered on its Zariski support. The term *width* refers to the number of successive Cartier layers in a jet.

2 Birational preparation

The purpose of this section is to replace the hypothetical case of maximal nef dimension by a smooth threefold carrying a single connected nef cycle. The torsion assertion below concerns the entire Cartier cycle, with its nilpotents.

Proposition 2.1 (The prepared cycle). *Suppose that k is uncountable and algebraically closed of characteristic $p > 3$, and that X is a projective terminal threefold with K_X nef, $\nu(K_X) = 1$, and $n(X, K_X) = 3$. There exist a smooth projective threefold V , a generically finite separable morphism $\rho : V \rightarrow X$, a very ample divisor H on V , and a nonzero effective Cartier divisor*

$$D = \sum_{\alpha} m_{\alpha} D_{\alpha}, \quad m_{\alpha} \in \mathbb{Z}_{>0}, \quad \gcd_{\alpha} m_{\alpha} = 1,$$

with the following properties.

- (i) *The support of D is connected and has simple normal crossings. The divisor D is nef, and $D|_{D_{\alpha}}$ is numerically trivial for every α .*
- (ii) *There is no D -trivial curve through a very general point of V . Moreover $D \cdot H^2 > 0$ and $D^2 \cdot H = 0$.*
- (iii) *The invertible sheaf $\mathcal{O}_D(D)$ is torsion. Its order will be denoted by r .*
- (iv) *For an integer $b \geq 0$, an open neighborhood U of $|D|$, and integers b_{α} , there is an isomorphism*

$$\omega_V^{\otimes p^b}|_U \simeq \mathcal{O}_U(B_*), \quad B_* = \sum_{\alpha} b_{\alpha} D_{\alpha}. \quad (2.1)$$

Consequently $K_V \cdot D \cdot A = 0$ for every Cartier divisor A on V .

The case $n(X, K_X) \leq 2$ is already covered by [32, Theorem 1.8], so this is the remaining nef-dimension case. The rest of this section proves Proposition 2.1. Starting with an effective pluricanonical divisor, we isolate a reduced boundary component by the relative minimal model program. Surface adjunction supplies torsion on its reduction; local cohomology and p -power gluing then give torsion on the full Cartier cycle. A tame cover and primitive division produce (V, D) , and numerical proportionality transfers the maximal-nef-dimension condition to D . The intervening local surface arguments establish regularity in codimension two and justify reduced-boundary adjunction. The use of a connected primitive cycle follows the canonical-divisor viewpoint of Miyaoka [25, Definition 2.4 and Lemma 2.5]; the torsion and covering arguments here take place in characteristic p .

2.1 Resolution and field extension

We use projective resolution in dimension three [5, 6, 7]. The projective form that is unchanged over the regular locus is stated explicitly in [3, p. 1, introductory theorem]. It applies to our quasiprojective threefolds over the perfect field k . For divisor supports on a regular threefold we use embedded resolution [4, Corollary 0.4]: a reduced closed subscheme of dimension at most two in a regular excellent scheme admits a projective modification, unchanged off that subscheme, whose reduced inverse image is snc. The resolutions may be chosen to be isomorphisms over the specified regular open set. In particular, pairs admit the log resolutions used below.

We also use the discrepancy criterion for a simple normal crossing subpair on a smooth variety, allowing negative coefficients: coefficients at most one give log canonicity, and coefficients strictly less than one give the klt property. One can see the relevant inequalities directly. At the generic point of a center, use logarithmic differentials for the components through it and regular differentials for the other coordinates. At the generic point of a divisor on a smooth model, with uniformizer t , each pulled-back logarithmic differential is a regular differential plus a multiple of dt/t . Their exterior product has at most a simple pole: two occurrences of dt/t have zero exterior product. The relative Jacobian formula gives the discrepancy inequality. Lowering a coefficient adds its positive difference times the order of that component. For an exceptional divisor whose center misses the boundary, the relative Jacobian has positive order. Indeed a birational morphism between smooth models cannot be generically unramified along an exceptional divisor, since it would then be quasi-finite there. These observations also explain why it suffices to compute thresholds on a log resolution.

We may replace k by an uncountable algebraically closed extension. Here are the descent and preservation points needed for this reduction. A normal variety over an algebraically closed field is geometrically normal and geometrically integral. In the present setting, geometric normality also follows from Serre's criterion: the field is perfect, residue fields of finite type are separably generated, and the relevant flat base-change maps have regular fibers. A smooth projective resolution remains smooth and integral after extension. The canonical rational line bundle base-changes, since its restriction to the smooth open set does, and reflexive extension determines it elsewhere.

On a resolution $g : T \rightarrow X$, terminality gives

$$K_T = g^* K_X + \sum_E a_E E, \quad a_E > 0.$$

The base-changed resolution has the same positive exceptional discrepancies. A further exceptional divisor has a positive relative discrepancy over the smooth model and nonnegative contributions from the effective divisor $\sum_E a_E E$; hence terminality is preserved. Nefness is preserved, for example by writing a nef class as a limit of ample rational classes, and intersection numbers preserve the numerical dimension. Finally, for every Cartier multiple qK_X ,

$$H^0(X_{k'}, \mathcal{O}_{X_{k'}}(qK_{X_{k'}})) = H^0(X, \mathcal{O}_X(qK_X)) \otimes_k k'.$$

Thus pluricanonical sections descend in the required sense of equality of dimensions. If necessary, take a further multiple divisible by the original Cartier index; the powers of two linearly independent sections remain linearly independent, since their nonconstant ratio cannot acquire a constant positive power in a function field with algebraically closed constant field. We do not assume that $\mathbb{Q}$ -factoriality is preserved by this extension: the modifications below supply it when needed.

2.2 The maximal nef-dimension branch

Nonvanishing [32, Theorem 1.4] supplies an effective Cartier divisor

$$0 \neq N_0 \sim mK_X, \quad m > 0.$$

It is nonzero because $K_X \cdot H_X^2 > 0$ for an ample divisor H_X . It is nef, and $N_0^2 \cdot H_X = 0$. Every prime component T of N_0 satisfies

$$N_0 \cdot H_X \cdot T = 0,$$

because these nonnegative numbers have a positive weighted sum equal to zero. In fact $N_0|_T \equiv 0$. To verify this for an integral curve $C \subset T$, choose a sufficiently large multiple of $H_X|_T$ with an effective Cartier member containing C . Its intersection with the nef restriction is zero; all components contribute nonnegatively, so the contribution of C is zero.

The nef reduction exists over our uncountable algebraically closed field [33, Definition 2.1 and Theorem 2.2]. If $n(X, K_X) \leq 2$, the low-nef-dimension abundance theorem [32, Theorem 1.8] makes K_X semiample. Since its numerical dimension is one, some Cartier multiple has at least two sections. We may therefore assume $n(X, K_X) = 3$. The defining curve property of the nef reduction says in this case that there is no N_0 -trivial curve through a very general point of X . Here *very general* means outside a countable union of proper closed subsets.

2.3 Isolating a boundary component

We use the relative threefold MMP in characteristic greater than three, including preservation of $\mathbb{Q}$ -factorial dlt pairs for the boundary driving the MMP, crepant dlt modifications, and the partial MMP with nef restored boundary [33, Theorem 2.20, Proposition 2.21, Definition 2.23, Lemma 2.24, and Theorem 2.27]. The characteristic-five case is included through [11]; the corresponding results in characteristic greater than five include [16]. Termination for the pseudoeffective lc divisor we use is [32, Theorem 1.5]. The comparisons of pullbacks use the negativity lemma [21, Lemma 3.39].

We recall why the dlt preservation used here concerns negative steps. On a common resolution of a negative divisorial contraction or flip, the old log pullback minus the new one is effective by negativity. On curves over the contraction target its degree is nonpositive, and is strictly negative if the curve has a nonconstant image on a modified side. A component of a projective fiber not contained in this effective divisor cannot meet it: its effective restriction is anti-nef, hence zero. Connectedness of fibers then makes any fiber meeting the support lie entirely in it. Fibers over the modification locus do meet the support, by the strict degree inequality. Thus discrepancies strictly increase for valuations centered in that locus. A zero log discrepancy on the output of a dlt step must consequently come from the unchanged open set, where the required simple normal crossing behavior persists. We will apply this to the lowered boundary. For a larger boundary whose log divisor is trivial along the steps we retain log canonicity by crepancy, and later take a new dlt modification.

Let $\lambda > 0$ be the log canonical threshold of N_0 on X . It is finite and rational, and is attained: on a log resolution it is the minimum of finitely many positive rational discrepancy-to-order ratios. Thus $(X, \lambda N_0)$ is lc and is not klt. Take a crepant projective $\mathbb{Q}$ -factorial dlt modification

$$\mu : (W, B) \longrightarrow (X, \lambda N_0).$$

All μ -exceptional divisors occur in B with coefficient one. One construction starts on a log resolution with the strict boundary and all exceptional divisors reduced. Its log divisor is the desired pullback plus an effective exceptional discrepancy divisor. The relative dlt MMP

removes that difference by negativity, leaving precisely the crepant divisors. This is the modification theorem cited above.

Put $N = \mu^* N_0$ and $\tau = m^{-1} + \lambda > 0$. Then

$$K_W + B \sim_{\mathbb{Q}} \tau N, \quad \text{Supp } B = \text{Supp } N. \quad (2.2)$$

Indeed an extracted log-canonical place must have center in $\text{Supp } N_0$, since X is klt away from that divisor. Its order on N_0 is positive. The modification is an isomorphism over the smooth open set $X_{\text{reg}} \setminus \text{Supp } N_0$: there it has no exceptional divisors, and a projective small birational morphism to a locally factorial variety is an isomorphism. For the latter assertion, push down a relatively ample Cartier divisor. A multiple of its pushdown is Cartier, and its pullback agrees with that multiple upstairs because there are no exceptional divisors. It therefore has degree zero on any contracted curve, contradicting relative ampleness unless there are no such curves.

Since (W, B) is dlt and not klt, choose a coefficient-one component S . Choose a rational $\epsilon > 0$ so small that both

$$B - \epsilon S \geq 0, \quad \tau N - \epsilon S \geq 0.$$

Run a $(K_W + B - \epsilon S)$ -MMP which is $(K_W + B)$ -trivial. In the partial-MMP notation the boundary is $B - \epsilon S$, and the restoration divisor is ϵS . The lowered pair is dlt and its log divisor is pseudoeffective by the displayed effective representative. Termination therefore applies. There is no Mori fiber output, since a pseudoeffective divisor cannot be anti-ample on the covering family of curves in a Mori fiber space.

We spell out what happens to N . On a common resolution of each step, its two pullbacks agree. Their difference is exceptional over the new model. A curve contracted over that model maps on the old model to a curve over the step's contraction target, and has N -degree zero, because the extremal ray is N -trivial. The difference is therefore numerically trivial over the new model, and is zero by negativity applied to it and to its negative. This proves the pullback assertion for flips as well as divisorial contractions.

It follows that the transform of N remains nef and numerically trivial on every component of its support. For the latter assertion, lift a curve on such a component to the common resolution. Its image on the old side is a point or a curve in the old support, where N is numerically trivial. The same pullback equality shows that the full pair in (2.2) remains crepant and lc. The component S is not contracted: every contracted ray R satisfies

$$S \cdot R > 0,$$

whereas an exceptional divisor of a divisorial contraction has negative degree on some contracted curve, again by negativity. Moreover a negative curve for the effective divisor $\tau N - \epsilon S$ lies in its support. Thus all steps are isomorphisms off the support under consideration.

Write Y, B_Y, S_Y, N_Y for the output and its transforms. The partial-MMP conclusion gives

$$K_Y + B_Y \sim_{\mathbb{Q}} \tau N_Y, \quad K_Y + B_Y - \delta S_Y \text{ nef} \quad (2.3)$$

for some rational $\delta > 0$. The variety Y is $\mathbb{Q}$ -factorial and (Y, B_Y) is lc. If $T \neq S_Y$ is a component of N_Y , then $-S_Y|_T$ is nef by (2.3) and $N_Y|_T \equiv 0$. But $S_Y|_T$ is effective and $\mathbb{Q}$ -Cartier. If nonzero, it has positive degree against an ample divisor on the integral projective surface T , contradicting anti-nefness. Consequently

$$S_Y \cap T = \emptyset \quad (T \neq S_Y).$$

The pair (Y, S_Y) is lc, and, with $s = \text{mult}_{S_Y} N_Y > 0$,

$$S_Y|_{S_Y} \equiv 0, \quad K_Y + S_Y \sim_{\mathbb{Q}} \tau s S_Y \quad \text{on a neighborhood of } S_Y. \quad (2.4)$$

Take a new crepant projective $\mathbb{Q}$ -factorial dlt modification

$$f : (Y_d, \Sigma) \longrightarrow (Y, S_Y),$$

where Σ is the strict transform of S_Y together with the exceptional divisors, all reduced. We claim

$$\text{Supp } f^* S_Y = \text{Supp } \Sigma. \quad (2.5)$$

For an extracted divisor E , let a_E denote log discrepancy. Log canonicity gives

$$0 \leq a_E(Y, B_Y) = a_E(Y, S_Y) - \text{ord}_E(B_Y - S_Y) = -\text{ord}_E(B_Y - S_Y).$$

Both terms on the right must vanish. Crepancy with $(X, \lambda N_0)$ then gives $a_E(X, \lambda N_0) = 0$; since X is klt, $\text{ord}_E N_0 > 0$. The pullback equalities just proved give $\text{ord}_E N_Y > 0$. The other components of N_Y have the same support as $B_Y - S_Y$ and order zero at E , so $\text{ord}_E S_Y > 0$. This proves (2.5). The morphism f is an isomorphism off S_Y , by the small-morphism argument and $\mathbb{Q}$ -factoriality of Y . Near Σ we therefore have

$$K_{Y_d} + \Sigma \sim_{\mathbb{Q}} \tau s f^* S_Y, \quad f^* S_Y|_{\Sigma} \equiv 0. \quad (2.6)$$

2.4 Local surface intersections and terminal regularity

The adjunction calculation for Σ uses an intersection inequality on excellent local surfaces, whose residue fields need not be perfect. We prove that inequality first and also derive regularity of X in codimension two. For a resolution unchanged over $X_{\text{reg}} \setminus \text{Supp } N_0$, this regularity ensures that exceptional divisors with centers outside $\text{Supp } N_0$ are disjoint from its total pullback.

Lemma 2.2 (Exceptional surface intersections). *Let P be an excellent normal local surface with closed residue field l , and let $g : T \rightarrow P$ be a nontrivial projective regular resolution which is an isomorphism away from the closed point. Write G_i for its exceptional integral curves, and take intersection degrees over l . Then $G_i^2 < 0$ for every i . More generally, if an exceptional real divisor $Z = \sum_i z_i G_i$ has a positive coefficient, there is an index j with $z_j > 0$ and $Z \cdot G_j < 0$.*

Proof. The exceptional fiber is connected, by proper birationality and normality of P . Choose a nonzero nonunit $t \in \mathcal{O}_P$ and write

$$\text{div}_T(t) = A + \sum_i c_i G_i, \quad c_i > 0,$$

where A is the strict-transform part. It meets the exceptional fiber: the strict transform of a component of the divisor of t has a point over the closed point by properness. For $C = \sum_i c_i G_i$ we have

$$C \cdot G_i = -A \cdot G_i \leq 0.$$

Distinct curves have nonnegative intersections. Connectedness therefore shows $G_i^2 < 0$: every vertex has either a neighbor or, if it is the only vertex, a positive intersection with A .

For the second assertion put $a = \max_i (z_i/c_i) > 0$, and let I be the nonempty set where this maximum is attained. Choose $j \in I$ which either meets A or has a neighbor outside I . Such a vertex exists by connectedness and the fact that A meets the fiber. Then

$$Z \cdot G_j = aC \cdot G_j + \sum_i (z_i - ac_i) G_i \cdot G_j < 0.$$

The term with $i = j$ is zero; all other terms are nonpositive, and the chosen vertex makes at least one term strictly negative. $\square$

Lemma 2.3 (Terminal threefolds are regular in codimension two). *A terminal threefold in the present setting has only isolated singularities.*

Proof. Localize at a codimension-two point. This gives an excellent normal local surface P , with residue field l , and a projective regular resolution $g : T \rightarrow P$ obtained by localizing a threefold resolution. The residue field l need not be perfect. We keep all intersection degrees over l .

For an integral exceptional curve G , divisor adjunction and curve duality give

$$(K_T + G) \cdot G = 2 \dim_l H^1(G, \mathcal{O}_G) - 2 \dim_l H^0(G, \mathcal{O}_G). \quad (2.7)$$

To specify the duality in this formula, use the dualizing complex on P localized from the threefold and its exceptional inverse image on T . Transitivity for $G \rightarrow T \rightarrow P$ and $G \rightarrow \operatorname{Spec} l \rightarrow P$ identifies the adjunction sheaf on G with its dualizing sheaf over l , up to tensoring with a one-dimensional l -vector space. This does not change degree. Proper duality and the formula $\chi(\mathcal{L}) - \chi(\mathcal{O}_G) = \deg_l \mathcal{L}$ prove (2.7) [13, 15].

Suppose $K_T \cdot G < 0$, and set $l' = H^0(G, \mathcal{O}_G)$ and $d = [l' : l]$. Both G^2 and $K_T \cdot G$ are negative multiples of d . Equation (2.7) and Lemma 2.2 force

$$H^1(G, \mathcal{O}_G) = 0, \quad G^2 = K_T \cdot G = -d.$$

The line bundle $\mathcal{O}_G(-G)$ consequently has degree one over l' . Riemann–Roch gives at least two independent sections over l' . Every nonzero section has a zero scheme of length one. Two independent sections have no common zero: otherwise their effective Cartier divisors coincide, so their ratio is a global unit, an element of $l'^\times$. They define a nonconstant map $G \rightarrow \mathbb{P}_{l'}^1$ of degree one. It is finite, and a finite birational morphism to a normal curve is an isomorphism. Hence

$$G \simeq \mathbb{P}_{l'}^1, \quad \mathcal{O}_G(G) \simeq \mathcal{O}_{\mathbb{P}_{l'}^1}(-1).$$

For completeness this curve can be contracted while retaining a projective regular resolution over P . Choose a relatively very ample Cartier divisor A , put $a = \deg_{l'}(A|_G) > 0$, and set $Q = A + aG$. The divisor $Q - G$ has positive degree on every component of the closed fiber, and is relatively ample by the fiber criterion for ampleness [10]. The restriction $Q|_G$ is trivial. For every $n \geq 1$, a trivializing section of $\mathcal{O}_G(nQ)$ extends successively to nG , because the obstruction groups are

$$H^1(G, \mathcal{O}_G(nQ - jG)) = H^1(\mathbb{P}_{l'}^1, \mathcal{O}(j)) = 0 \quad (j \geq 1).$$

For large n , relative Serre vanishing applied to $n(Q - G)$ lifts these sections from nG to T . They generate along G . Away from G , the sections of $n(Q - G)$, multiplied by the canonical section of nG , generate and separate points and tangent directions. Thus nQ defines a projective contraction whose only nontrivial fiber is G . Its Stein factorization has normal target and is an isomorphism off G .

Let q be the image of G . Formal functions identifies the completed local ring at q with

$$\varprojlim_n H^0(nG, \mathcal{O}_{nG}).$$

The G -adic and inverse-image q -adic neighborhoods are cofinal, since they have the same support. The successive kernels have graded pieces

$$H^0(G, \mathcal{O}_G(-nG)) = H^0(\mathbb{P}_{l'}^1, \mathcal{O}(n)).$$

There is no obstruction in H^1 , so the associated graded algebra is the polynomial ring $l'[x, y]$. More explicitly, let F_n be the kernel of the map from the completed local ring to $H^0(nG, \mathcal{O}_{nG})$.

Then F_1 is its maximal ideal, and the F_n -filtration is cofinal with the maximal-ideal-adic filtration. Choose lifts of the two degree-one generators. Generation of the graded algebra gives $F_1 \subset (x, y) + F_n$ for every n , by successive approximation, using arbitrary lifts of coefficients from l' . Finitely generated ideals in the complete Noetherian local ring are closed, so $F_1 = (x, y)$. No embedding of the residue field l' into the completed local ring is required. Its dimension is two; hence it, and then the local ring at q , is regular. This proves the claimed contraction using formal functions and Serre vanishing [15, III, Sections 5 and 11].

There are only finitely many exceptional curves. Contracting a K_T -negative one whenever possible therefore yields a regular resolution with K_T relatively nef. If any exceptional curves remain, terminality gives

$$E = K_T - g^*K_P = \sum_i a_i G_i, \quad a_i > 0.$$

The discrepancies of surviving curves have not changed, since they are read at their generic points. Lemma 2.2 gives $E \cdot G_j < 0$ for some j , contradicting relative nefness. There are no exceptional curves left, and the proper birational map is an isomorphism. Thus the codimension-two point was regular. Normality already gives regularity in codimension one, so the remaining singular locus of the projective threefold is finite. $\square$

2.5 Adjunction on the reduced boundary

We return to the dlt pair (Y_d, Σ) satisfying (2.6). To apply surface abundance, we must identify its restricted log canonical class with that of a projective slc surface.

Let $\sigma : \Sigma' \rightarrow \Sigma$ be the finite S_2 -ification. It is an isomorphism away from finitely many closed points [33, Proposition 2.14]. Reduced-boundary adjunction gives

$$(K_{Y_d} + \Sigma)|_{\Sigma'} = K_{\Sigma'} + \Delta', \quad (2.8)$$

where (Σ', Δ') is an slc surface pair [33, Lemma 2.19]. Restrictions here and below mean rational line bundles, or sufficiently divisible Cartier multiples. We give the local verification needed in this dlt situation.

Choose a log resolution and write

$$K_{\tilde{Y}} + \tilde{\Sigma} + \sum_i \beta_i E_i = h^*(K_{Y_d} + \Sigma), \quad \beta_i \leq 1. \quad (2.9)$$

At a codimension-one point of Σ , localize the ambient threefold to a normal local surface P , and write $h : T \rightarrow P$ for the localized regular resolution. If the pair is not snc at this point, dlt gives $\beta_i < 1$ on the exceptional curves over it. Set

$$J = -\sum_i \lfloor \beta_i \rfloor E_i, \quad M = J - \tilde{\Sigma}, \quad A_{\text{fr}} = \sum_i \{\beta_i\} E_i.$$

Here J is effective, and

$$M - K_T - A_{\text{fr}} \equiv_P 0. \quad (2.10)$$

We claim $R^1 h_* \mathcal{O}_T(M) = 0$. For every nonzero effective integral exceptional cycle Z , the divisor $Z - A_{\text{fr}}$ has a positive coefficient. Lemma 2.2 gives a component E_j with positive coefficient in Z such that $(Z - A_{\text{fr}}) \cdot E_j < 0$. Using (2.10),

$$(M - Z + E_j) \cdot E_j > (K_T + E_j) \cdot E_j = \deg_l \omega_{E_j/l}.$$

Curve duality gives $H^1(E_j, \mathcal{O}_{E_j}(M - Z + E_j)) = 0$: the dual space consists of sections of a line bundle of negative degree. The exact sequence

$$0 \longrightarrow \mathcal{O}_{E_j}(M - Z + E_j) \longrightarrow \mathcal{O}_Z(M) \longrightarrow \mathcal{O}_{Z-E_j}(M) \longrightarrow 0$$

then proves $H^1(Z, \mathcal{O}_Z(M)) = 0$ by induction on the sum of the coefficients of Z . Exceptional cycles give a cofinal system of neighborhoods of the exceptional fiber. Formal functions, followed by faithful completion for the coherent higher direct image, proves the claim.

Normality gives $h_*\mathcal{O}_T(J) = \mathcal{O}_P$: a rational function with possible poles only along exceptional divisors is regular at every codimension-one point of P , hence regular on P . The preceding vanishing and

$$0 \longrightarrow \mathcal{O}_T(M) \longrightarrow \mathcal{O}_T(J) \longrightarrow \mathcal{O}_{\tilde{\Sigma}}(J) \longrightarrow 0$$

give a surjection $\mathcal{O}_P \rightarrow h_*\mathcal{O}_{\tilde{\Sigma}}(J)$. It factors through $h_*\mathcal{O}_{\tilde{\Sigma}}$, which is a subsheaf of the target. The strict boundary has no exceptional component, so its proper map to the local boundary is quasi-finite and hence finite. The kernel of $\mathcal{O}_P \rightarrow h_*\mathcal{O}_{\tilde{\Sigma}}$ is precisely the ideal of the reduced boundary. The surjection thus identifies $\tilde{\Sigma}$ with that boundary and also gives $\mathcal{O}_{\tilde{\Sigma}} = \mathcal{O}_{\tilde{\Sigma}}(J)$ under their natural inclusion. In particular J misses the strict boundary.

Two strict branches cannot cross above this non-snc point. In a nontrivial regular surface resolution any such intersection over the closed point lies on the exceptional fiber; the total divisor would then have three components through a point on a surface, contrary to snc. The trivial-resolution case would already be snc downstairs. Thus the boundary is regular at a non-snc point of codimension one, and every exceptional coefficient meeting its strict transform is nonnegative. At the snc points, the reduced boundary is nodal. This proves that Σ' is nodal in codimension one.

Here is the log-line identification in (2.8). Use rational residue forms to identify the plurilog-canonical line with pluricanonical forms on each boundary component. At snc points, Cartier divisor adjunction gives this identification, including along the nodes. At the other codimension-one points, restrict (2.9) to the smooth strict component. The preceding local calculation identifies it with the boundary there and shows that the resulting different is effective, since the exceptional coefficients meeting it are nonnegative. It has coefficient zero along the nodes. The identification extends by the S_2 property, using reflexive powers with the indicated divisorial twist. Its right side is therefore $\mathbb{Q}$ -Cartier with exactly the rational line bundle on the left side of (2.8).

On the normalization of Σ' , the boundary is the transform of Δ' together with the conductor, the latter with coefficient one. This is the residue formula at a node, extended from codimension one. Its log pullback to the smooth strict component $\tilde{\Sigma}_i$ is

$$K_{\tilde{\Sigma}_i} + \left((\tilde{\Sigma} - \tilde{\Sigma}_i) + \sum_j \beta_j E_j \right) |_{\tilde{\Sigma}_i}.$$

This follows from smooth adjunction applied to (2.9), with compatible rational residue forms, so it is an equality of log pullbacks, not just numerical classes. The displayed subpair is snc with every coefficient at most one. The discrepancy criterion at the start of the section makes it lc. Thus Σ' is S_2 , nodal in codimension one, and lc after normalization with conductor; this proves the asserted slc property.

By (2.6), the log canonical class in (2.8) is numerically trivial. The surface Σ' is projective, since it is finite over Σ . Abundance for projective slc surfaces in positive characteristic [26, Theorem 3.1] makes this class semiample, hence $\mathbb{Q}$ -linearly trivial. Indeed a globally generated numerically trivial line bundle gives a constant morphism on each connected component and is trivial there. It follows from (2.6), whose coefficient τs is positive rational, that some positive Cartier multiple L of f^*S_Y satisfies

$$\mathcal{O}_{Y_d}(L)|_{\Sigma \setminus Z} \simeq \mathcal{O}_{\Sigma \setminus Z} \quad (2.11)$$

for a finite set Z of closed points. Its support is Σ , and $L|_{\Sigma_i} \equiv 0$ for every component.

2.6 Torsion on the full Cartier cycle

We now pass from (2.11) to the actual nonreduced divisor L .

Lemma 2.4 (Extending torsion across finitely many points). *Let Y be a normal projective threefold over an algebraically closed field of characteristic $p > 0$, and let L be an effective Cartier divisor. Suppose that L is numerically trivial on every integral component of its support and that $\mathcal{O}_Y(L)|_{L_{\text{red}} \setminus Z}$ is trivial for a finite set of closed points Z . Then $\mathcal{O}_L(L)$ is torsion.*

Proof. Replacing Z by $Z \cap |L|$, we may assume $Z \subset |L|$. For $z \in Z$, put $R = \mathcal{O}_{Y,z}$ and let $\mathfrak{m}$ be its maximal ideal. Normality gives depth at least two, so

$$H_{\mathfrak{m}}^1(R) = 0.$$

Every nonmaximal localization of R is a normal local ring of dimension at most two, and is consequently Cohen–Macaulay. This implies that $H_{\mathfrak{m}}^2(R)$ has finite length. Explicitly, present $R = A/I$, where A is a regular local ring of dimension n , essentially of finite type over the field. The height of I is $n - 3$. At a nonmaximal prime in $\text{Spec } R$, catenarity and Auslander–Buchsbaum give projective dimension $n - 3$ over the corresponding localization of A . Hence $\text{Ext}_A^{n-2}(R, A)$ is supported only at the closed point. It is a finite module, thus has finite length. Local duality identifies $H_{\mathfrak{m}}^2(R)$ with its Matlis dual, up to completion, and proves the assertion [14].

Let t_z be a local equation of L . Choose j so large that t_z^{j-1} kills $H_{\mathfrak{m}}^2(R)$ for every $z \in Z$. There are only finitely many points, so the same j works at all of them. The restriction sequences fit into a commutative diagram

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathcal{O}_Y((u-j)L) & \rightarrow & \mathcal{O}_Y(uL) & \rightarrow & \mathcal{O}_{jL}(uL) \rightarrow 0 \\ & & \downarrow & & \parallel & & \downarrow \\ 0 & \rightarrow & \mathcal{O}_Y((u-1)L) & \rightarrow & \mathcal{O}_Y(uL) & \rightarrow & \mathcal{O}_L(uL) \rightarrow 0. \end{array}$$

In local trivializations the left vertical map is multiplication by t_z^{j-1} . The vanishing of the ambient H_Z^1 makes the connecting maps from the H_Z^1 of the rightmost terms to the ambient H_Z^2 injective. The choice of j therefore gives

$$H_Z^1(Y, \mathcal{O}_{jL}(uL)) \longrightarrow H_Z^1(Y, \mathcal{O}_L(uL)) \quad \text{equal to zero for every integer } u. \quad (2.12)$$

The assertion is uniform in u : twisting by uL only changes local trivializations of these finite-length modules.

Fix this j . The kernel of $\mathcal{O}_{jL \setminus Z} \rightarrow \mathcal{O}_{L_{\text{red}} \setminus Z}$ is nilpotent. Lift a chosen generator of the trivial line $\mathcal{O}_Y(L)|_{L_{\text{red}} \setminus Z}$ to local generating sections on a finite open cover of $jL \setminus Z$. On overlaps their differences are in the nilpotent ideal. For $u = p^e$ sufficiently large, their u -th powers agree, since the p^e -power map is additive in characteristic p and kills that ideal. The powers therefore give a trivializing section

$$s_j \in H^0(jL \setminus Z, \mathcal{O}_{jL}(uL)).$$

Project it to a section s on $L \setminus Z$. In the exact sequence of sections with support in Z , its obstruction to extending over L is the image of the obstruction for s_j . Equation (2.12) kills that image. Thus s extends to $H^0(L, \mathcal{O}_L(uL))$.

This extension is a generator everywhere. On each integral projective surface component T of L_{red} it is nonzero, since it is a generator off Z . If it vanished somewhere on T , its zero

scheme would be a nonempty effective Cartier divisor: a nonzero section of an invertible sheaf on an integral scheme is a non-zero-divisor in local trivializations. Such a divisor has positive degree against an ample class, whereas $\mathcal{O}_T(uL)$ is numerically trivial. This is impossible. No normality of T is needed for this intersection argument. Nonvanishing after reduction implies invertibility on L , by the local criterion for a unit modulo a nilpotent ideal. We conclude $\mathcal{O}_L(uL) \simeq \mathcal{O}_L$, as required. $\square$

Applying Lemma 2.4 to Y_d and L proves that $\mathcal{O}_L(L)$ is torsion.

2.7 A tame cover and the primitive cycle

Choose a smooth projective common resolution V_0 of Y_d and X . It may be chosen to be an isomorphism over the common open set

$$X_{\text{reg}} \setminus \text{Supp } N_0$$

and to resolve the relevant divisor supports. Let $\rho_0 : V_0 \rightarrow X$ be its map to X . The pullbacks of N_Y and N_0 agree on V_0 , by the pullback equalities through the MMP.

The section cutting out $N_0 \sim mK_X$ gives a rational section θ of $\omega_{V_0}^{\otimes m}$ with

$$\text{div}(\theta) = \rho_0^* N_0 + m(K_{V_0} - \rho_0^* K_X). \quad (2.13)$$

Every exceptional center outside $\text{Supp } N_0$ is an isolated singular point of X , by Lemma 2.3 and the choice of the resolution. The divisors over those centers are disjoint from $\rho_0^* N_0$. Thus the support in (2.13) consists of the total pullback support and possibly divisors disjoint from it.

Write

$$m = p^b m', \quad b \geq 0, \quad (m', p) = 1.$$

Choose a nonzero rational generator η of $\omega_{V_0}^{\otimes p^b}$, so that $\theta = a\eta^{m'}$ for $a \in k(V_0)^\times$. Adjoin one root v of $v^{m'} = a$, and normalize V_0 in the resulting field. The extension is finite separable, possibly of degree smaller than m' . Its normalization is projective. Away from $\text{Supp } \text{div}(\theta)$, this construction is an integral component of the torsor of m' -th roots of a nowhere-vanishing section of a line bundle. It is finite étale there, since m' is invertible in k .

Resolve this normalization, and then the total divisor support, keeping that regular étale open set unchanged. Denote the resulting smooth projective variety by V , and its generically finite separable map to V_0 by π . The root $v\eta$ is a rational section of $\pi^* \omega_{V_0}^{\otimes p^b}$. The determinant of the differential of π is a nonzero rational section of $\omega_V \otimes \pi^* \omega_{V_0}^{-1}$; it is nonzero because π is separable. Multiplying $v\eta$ by the p^b -th power of this Jacobian section produces a rational section of $\omega_V^{\otimes p^b}$. Its divisor is supported over $\text{Supp } \text{div}(\theta)$, since π is étale off that support. In particular it is supported on the total pullback of N_0 , together with divisors disjoint from it.

Let L_V be the pullback of L from Y_d , and choose one connected component C of $\text{Supp } L_V$. The isolation of S_Y shows that C is disjoint from the remainder of the total pullback support of N_0 . Write the part of L_V supported on C as

$$aD, \quad D = \sum_{\alpha} m_{\alpha} D_{\alpha}, \quad a = \gcd_{\alpha}(\text{mult}_{D_{\alpha}} L_V).$$

Then D is a nonzero primitive effective Cartier divisor, since V is smooth. Its support is connected and snc by the final resolution.

The pullback L_V is numerically trivial on its support: any curve there maps to a point or to a curve in Σ , where L is numerically trivial. On a neighborhood of C , $L_V = aD$. Consequently $D|_{D_{\alpha}} \equiv 0$. For a curve not contained in $\text{Supp } D$, effectivity gives nonnegative

intersection with D ; for one contained in a component the degree is zero. Hence D is nef. We have

$$D^2 \cdot H = 0, \quad D \cdot H^2 > 0$$

for every very ample divisor H .

If $\mathcal{O}_L(tL) \simeq \mathcal{O}_L$, its pullback is trivial on the Cartier scheme L_V . Near C , the inclusion $D \subset aD = L_V$ is an inclusion of closed subschemes. Restricting the pulled-back trivialization gives

$$\mathcal{O}_D(taD) \simeq \mathcal{O}_D.$$

Thus $\mathcal{O}_D(D)$ is torsion. This argument does not claim that its order is unchanged under the cover or primitive division.

The rational canonical section constructed above has, on a neighborhood of C , divisor supported on D ; all other components of its divisor are disjoint from C . It therefore gives (2.1). For any Cartier divisor A on V ,

$$p^b K_V \cdot D \cdot A = B_* \cdot D \cdot A = \sum_{\alpha} b_{\alpha} c_1(\mathcal{O}_{D_{\alpha}}(D)) \cdot A|_{D_{\alpha}} = 0.$$

This includes $K_V^2 \cdot D = 0$.

2.8 Numerical proportionality and the final interface

Put $\rho = \rho_0 \circ \pi$ and $N_{\text{tot}} = \rho^* N_0$. Both D and N_{tot} are nef and nonzero. The latter is numerically trivial on every component of its support: every curve there maps to a point or to a curve in $\text{Supp } N_0$. We have

$$D^2 \cdot H = N_{\text{tot}}^2 \cdot H = D \cdot N_{\text{tot}} \cdot H = 0, \quad D \cdot H^2 > 0, \quad N_{\text{tot}} \cdot H^2 > 0.$$

The surface Hodge index theorem implies that

$$D \equiv c N_{\text{tot}}, \quad c = \frac{D \cdot H^2}{N_{\text{tot}} \cdot H^2} > 0. \quad (2.14)$$

Here is a way to check the numerical equivalence on every curve rather than just on one general surface. Given an integral curve $C_0 \subset V$, take $a \gg 0$ and an integral surface $T \in |aH|$ containing C_0 . Such a surface exists by Bertini irreducibility with this prescribed base locus [17]: for sufficiently large a , generators of the twisted ideal of C_0 , multiplied by very ample sections, give a linear system with no fixed divisor and with a birational map to its image off C_0 . The image has dimension three, so a general member is integral.

On a smooth projective resolution $\tilde{T}$ of T , the pullbacks of D and N_{tot} are nef, isotropic, mutually orthogonal classes. The pullback of $H|_T$ has positive square. The Hodge index form on $N^1(\tilde{T})_{\mathbb{R}}$ has signature $(1, \dim_{\mathbb{R}} N^1(\tilde{T})_{\mathbb{R}} - 1)$ [15, V, Section 1]; its restriction to the orthogonal complement of any positive-square class is negative definite. Two nonzero orthogonal isotropic classes with positive intersection against that class are therefore positively proportional. Their ratio is the number c in (2.14), since intersection with the pullback of $H|_T$ gives $aD \cdot H^2$ and $aN_{\text{tot}} \cdot H^2$. Choose a curve on the resolution dominating C_0 ; projection of degrees proves $(D - cN_{\text{tot}}) \cdot C_0 = 0$. This proves (2.14).

Finally a very general point of V lies in the generically finite locus of ρ and maps outside the countable union excluded by the nef reduction on X . A curve through such a point has a curve as its image. If it were D -trivial, (2.14) and the projection formula would make its image N_0 -trivial, a contradiction. This establishes the remaining assertion of Proposition 2.1. The proof of that proposition is complete. In the subsequent sections only V, D, H , the torsion order r , and the local canonical relation (2.1) are retained; the auxiliary birational models are no longer needed.

3 Slopes, foliations, and polar parts

We retain the smooth projective threefold V and the cycle D of Proposition 2.1. Fix a very ample divisor H on V . The numerical properties used in this section are

$$H^2D > 0, \quad HD^2 = 0, \quad K_V DA = 0 \quad \text{for every divisor class } A. \quad (3.1)$$

We also use that D is nef and that no curve of D -degree zero passes through a very general point of V . The ground field is uncountable, algebraically closed, and of characteristic $p > 3$. We will obtain strong semistability and $c_2(V)D \geq 0$, followed by uniform bounds on vector-valued polar parts and the vanishing of scalar polar parts. The scalar vanishing starts the jet calculation in Section 4; the vector bounds control Laurent primitives in Section 5.

3.1 Slope calculus for the nef product

For a torsion-free coherent sheaf E of positive rank, set

$$\deg(E) = c_1(E)HD, \quad \mu(E) = \frac{\deg(E)}{\operatorname{rk}(E)}.$$

Stability and semistability will always refer to this slope. A sheaf is strongly semistable if all its iterated absolute Frobenius pullbacks are semistable. Although D need not be ample, the elementary slope calculus needed below still applies. We give the details that are relevant here.

An effective divisor has nonnegative degree, since H and D are nef. Saturating a subsheaf therefore does not decrease its degree. Every torsion-free sheaf E embeds in a finite direct sum of copies of $\mathcal{O}_V(aH)$ for some a : dualize a sufficiently positive global-generation map for $E^\vee$ and use $E \hookrightarrow E^{\vee\vee}$. If $B \subset E$ has rank b , taking exterior powers and then determinants gives a nonzero map from $\det B$ to some copy of $\mathcal{O}_V(baH)$. Hence

$$\mu(B) \leq aH^2D.$$

For ranks bounded by $\operatorname{rk} E$, the possible slopes form a discrete subset of $\mathbb{Q}$. Thus a maximal slope is attained. Among saturated subsheaves of maximal slope, take one of maximal rank. The usual sum-and-intersection argument shows that this subsheaf contains every other subsheaf of maximal slope. It is semistable, and the maximal slope of its torsion-free quotient is strictly smaller. Repeating on the quotient produces the Harder–Narasimhan filtration.

Write $\mu_{\max}(E)$ and $\mu_{\min}(E)$ for the largest and smallest slopes of its factors. Additivity of degree and the image of a morphism give the familiar implication

$$\mu_{\min}(E) > \mu_{\max}(F) \implies \operatorname{Hom}(E, F) = 0. \quad (3.2)$$

Indeed a nonzero image is both a torsion-free quotient of the source and a subsheaf of the target, which would force its slope to be at least the left-hand extreme and at most the right-hand extreme. Dualizing the filtration on the open set where its terms and factors are locally free gives

$$\mu_{\max}(E^\vee) = -\mu_{\min}(E), \quad \mu_{\min}(E^\vee) = -\mu_{\max}(E).$$

The omitted set has codimension at least two and does not change any determinant or degree. The same observation permits all ensuing bundle calculations on such open sets, with saturation or reflexive extension when returning to V .

Tensoring by a line bundle adds its degree to every slope. More generally, the same rule holds for a rank-one torsion-free factor after restriction to the open set where it is invertible. In particular, Equation (3.1) shows that twisting by iD leaves all slopes unchanged, for every $i \in \mathbb{Z}$. We will use this rank-one rule; no assertion about semistability of arbitrary tensor products is required.

3.2 Connections and distributions of rank at most two

Let $F_V: V \rightarrow V$ denote absolute Frobenius. A Frobenius pullback F_V^*E has its canonical connection, which in a pulled-back local frame differentiates the coefficients. We first record the elementary descent observation behind its use.

Lemma 3.1 (The connection detects destabilization). *Let E be a semistable torsion-free sheaf. If F_V^*E is unstable and B is its first Harder–Narasimhan term, then the connection induces a nonzero morphism on a big open set*

$$B \longrightarrow (F_V^*E/B) \otimes \Omega_V^1.$$

Proof. The induced map is $\mathcal{O}_V$ -linear by the Leibniz rule. Suppose it vanishes. Then the generic subspace defined by B is horizontal. Let $K = k(V)$ and choose a pulled-back basis of the ambient generic vector space. In a Grassmannian chart, represent this subspace by a matrix $(I \mid M)$. Horizontality says that every entry of M has zero differential. Since K is separably generated over the perfect field k , the kernel of $d: K \rightarrow \Omega_{K/k}^1$ is K^p . Taking p -th roots of the entries of M therefore defines a subspace before Frobenius pullback.

Saturate this subspace to a subsheaf $B_0 \subset E$. At every codimension-one point the relevant torsion-free quotients are free over the discrete valuation ring. Frobenius is flat on the smooth variety, so $F_V^*B_0$ is saturated there and agrees with B there: they have the same generic subspace. Consequently

$$\deg(B) = p \deg(B_0), \quad \mu(B_0) = \frac{\mu(B)}{p} > \frac{\mu(F_V^*E)}{p} = \mu(E),$$

contradicting semistability of E . □

The general connection criterion is also stated in [23, Corollary 2.4]. The argument above makes explicit why it applies to the present nef slope.

Lemma 3.2 (The first destabilizing term is a foliation). *If T_V is unstable, its first Harder–Narasimhan term A is closed under Lie brackets and under p -th powers of derivations.*

This rank-one and rank-two argument follows the foliation criterion in [31, Lemma 2.9]; we give the slope calculation for the nef product used here.

Proof. Put $Q = T_V/A$ and $a = \mu(A)$. By Equation (3.1), $\mu(T_V) = 0$, so

$$a > 0, \quad \mu_{\max}(Q) < a.$$

The rank of A is one or two. Work where A and Q are bundles. The bracket modulo A is an $\mathcal{O}_V$ -linear map $\wedge^2 A \rightarrow Q$. It vanishes automatically in rank one. In rank two, Q has rank one and slope $-2a$, whereas $\wedge^2 A$ has slope $2a$; Equation (3.2) again makes the map zero.

Once bracket closure holds, the p -th power of derivations modulo A defines an $\mathcal{O}_V$ -linear map

$$F_V^*A \longrightarrow Q. \tag{3.3}$$

For completeness, the two restricted-Lie identities needed for this assertion can be seen directly. For local derivations $u, v \in A$, set $w = tu + v$ with t a formal scalar. In characteristic p ,

$$\frac{d}{dt}w^p = \sum_{j=0}^{p-1} w^j u w^{p-1-j} = \text{ad}(w)^{p-1}(u).$$

The last equality follows from $\binom{p-1}{j} \equiv (-1)^j \pmod{p}$. Every coefficient of the right-hand side lies in the bracket-closed module A . The mixed coefficients of $w^p - t^p u^p - v^p$, whose degrees in t lie between 1 and $p-1$, thus belong to A . This proves additivity modulo A .

For a local function f , expanding $(fu)^p$ as a differential operator gives

$$(fu)^p - f^p u^p = \sum_{j=1}^{p-1} c_j u^j.$$

The left side is a derivation. At the generic point, if $u \neq 0$, choose x with $u(x) \neq 0$. For a largest index $j \geq 2$ with $c_j \neq 0$, the j -fold commutator with multiplication by x is $j!c_j(u(x))^j \neq 0$, while it must vanish for a derivation. Hence only the $j = 1$ term can remain. The scalar discrepancy is therefore a multiple of u , proving Frobenius linearity modulo A and hence Equation (3.3).

If $\text{rk } A = 1$, its source has slope $pa > a > \mu_{\max}(Q)$ and the map is zero. Suppose $\text{rk } A = 2$. In this case

$$\mu(Q) = -2a, \quad \mu_{\max}(\Omega_V^1) = 2a.$$

If $F_V^* A$ is semistable, its minimum slope is $pa > -2a$. Otherwise let L be its first Harder–Narasimhan term and let $R = F_V^* A/L$; both have rank one. Lemma 3.1 supplies a nonzero map

$$L \longrightarrow R \otimes \Omega_V^1.$$

Since R has rank one, the slope rules give

$$\mu(L) - \mu(R) \leq 2a, \quad \mu(L) + \mu(R) = 2pa.$$

It follows that

$$\mu_{\min}(F_V^* A) = \mu(R) \geq pa - a > -2a.$$

Thus Equation (3.2) makes Equation (3.3) zero in either case. $\square$

3.3 The inseparable quotient and its canonical divisor

We spell out the quotient construction to keep track of the sign in the canonical formula. These constructions and their use for tangent semistability also appear in [31, Section 2].

Lemma 3.3 (Quotient by the distribution). *Let $A \subset T_V$ be a saturated bracket-closed and p -closed subsheaf of rank d . There is a finite purely inseparable map $\pi: V \rightarrow W$ to a normal projective variety, with a factorization*

$$V \xrightarrow{\pi} W \xrightarrow{\rho} V^{(1)}$$

of relative Frobenius. On open sets whose complements have codimension at least two, the canonical bundles satisfy

$$\omega_V \simeq \pi^* \omega_W \otimes (\det A)^{\otimes (p-1)}. \quad (3.4)$$

Proof. On an affine open with ring R , let

$$P = \{f \in R : u(f) = 0 \text{ for every local section } u \in A\}.$$

The ring P lies between R^p and R . Since R is finite over R^p , P is finite over R^p and R is finite over P . These invariant rings glue to the asserted factorization of Frobenius; in particular both maps are finite and their underlying maps of topological spaces are homeomorphisms. The ring P is normal. Indeed, an element of $\text{Frac}(P)$ integral over P is integral over R , hence

belongs to the normal ring R , and all the derivations annihilate it. Thus it belongs to P . The variety W is projective because it is finite over the projective variety $V^{(1)}$.

Remove the codimension-at-least-two loci where A or T_V/A fails to be locally free. Near any remaining point choose functions $x_1, \dots, x_d$ whose differentials restrict to a basis of $A^\vee$, and choose the dual local derivations $u_1, \dots, u_d$. Thus $u_i(x_j) = \delta_{ij}$. Bracket closure implies $[u_i, u_j] \in A$; this bracket annihilates every x_l and is therefore zero. Likewise $u_i^p \in A$ annihilates all the x_l , so $u_i^p = 0$.

These commuting nilpotent derivations give an explicit invariant-ring presentation. For one derivation u with $u(x) = 1$, the coefficient $(p-1)!^{-1}u^{p-1}(f)$ lies in its kernel. Subtract its product with x^{p-1} and repeat with successively smaller powers. This writes every f uniquely as a polynomial of degree less than p in x , with coefficients in the kernel of u . Applying this procedure successively to the commuting u_i gives

$$R = \bigoplus_{0 \leq I_1, \dots, I_d < p} P x^I \simeq P[X_1, \dots, X_d]/(X_1^p - c_1, \dots, X_d^p - c_d), \quad c_i = x_i^p \in P. \quad (3.5)$$

Uniqueness follows by applying the same successive highest derivatives to a putative relation. In particular, π is locally free of degree p^d on this open set.

We may further remove the singular locus of W and its inverse image; normality and finiteness ensure that these removed sets also have codimension at least two. The presentation in Equation (3.5) now yields the exact sequence of vector bundles

$$0 \longrightarrow F_V^*(A^\vee) \longrightarrow \pi^* \Omega_W^1 \longrightarrow \Omega_V^1 \longrightarrow A^\vee \longrightarrow 0. \quad (3.6)$$

Here the last map is evaluation on A . To verify local freeness of the first image, put $N = \langle dc_1, \dots, dc_d \rangle$ inside $R \otimes_P \Omega_{P/k}^1$. The presentation gives

$$\Omega_{R/k}^1 \simeq ((R \otimes_P \Omega_{P/k}^1)/N) \oplus \bigoplus_{i=1}^d R dx_i,$$

because the relations have no differential in the X_i directions. Smoothness of R makes the first summand locally free of rank $3 - d$. Smoothness of P then makes N a locally free direct summand of rank d in the rank-three bundle $R \otimes_P \Omega_{P/k}^1$. The displayed d generators of N are a basis: locally their surjection from R^d onto a free module of rank d is an isomorphism. This proves the required exactness at every point of the chosen open set.

To identify its transition maps, let x'_j be another such set of functions and write it in the basis x^I over P . Taking p -th powers and differentiating downstairs shows that, after pullback, the coefficient of dc_l in $d((x'_j)^p)$ is $u_l(x'_j)^p$. On the other hand, the coefficient of the l -th basis vector of $A^\vee$ in $dx'_j|_A$ is $u_l(x'_j)$. The first term of Equation (3.6) is therefore precisely $F_V^*(A^\vee)$. Taking determinants in that sequence gives

$$\pi^* \omega_W \simeq (\det A)^{-p} \otimes \omega_V \otimes \det A,$$

which is Equation (3.4). □

3.4 Strong semistability of the tangent bundle

Proposition 3.4. *The tangent bundle T_V is strongly semistable for the slope $c_1(-)HD/\mathrm{rk}(-)$. Moreover,*

$$c_2(V)D \geq 0. \quad (3.7)$$

Proof. We first prove semistability by the foliation and rational-curve argument of [31, Lemma 2.10], with the quantitative bend-and-break statement specified below. Suppose that A is the first destabilizing term, with $a = \mu(A) > 0$ and $d = \text{rk } A$. By Lemmas 3.2 and 3.3, it has a finite purely inseparable quotient $\pi: V \rightarrow W$. On the big smooth open set where Equation (3.4) holds, put $\Lambda = -\pi^*K_W$. Its extension as a divisor class on V is

$$\Lambda = (p-1)c_1(A) - K_V, \quad \Lambda HD = (p-1)da > 0.$$

No global Cartier property of K_W is needed: its degree will be taken only on curves contained in this smooth open set.

Let $H^{(1)}$ and $D^{(1)}$ denote the corresponding divisor classes on $V^{(1)}$, and set

$$H' = \rho^*H^{(1)}, \quad D' = \rho^*D^{(1)}.$$

Then H' is ample and Cartier, D' is nef and Cartier, and

$$\pi^*H' = pH, \quad \pi^*D' = pD.$$

For a positive integer t , take sufficiently divisible positive multiples of H' and $H' + tD'$ which are very ample. Their general complete intersection is a smooth projective integral curve C_t wholly contained in the open set under consideration. Here is the avoidance point in this assertion. The complement has dimension at most one. A general first divisor meets it in finitely many points and contains none of its curve components; a general second divisor avoids those finitely many points. Bertini on the smooth open set, and irreducibility for the successive very ample sections, give the required curve.

Projection formula, computed where π is locally free, cancels both the degree of π and the chosen complete-intersection multiples from the degree ratio. It gives

$$\begin{aligned} \frac{D'C_t}{-K_W C_t} &= \frac{pD H(H+tD)}{\Lambda H(H+tD)} \\ &= \frac{pDH^2}{\Lambda H^2 + t\Lambda HD} \rightarrow 0 \quad (t \rightarrow \infty). \end{aligned} \tag{3.8}$$

The denominator is positive for all sufficiently large t .

We use the following quantitative form of bend-and-break [18, Theorem 1.1]: if Y is projective over an algebraically closed field, $C \subset Y_{\text{sm}}$ is an integral curve with $K_Y C < 0$, and N is a nef real Cartier divisor, then every closed point of C lies on a rational curve $\Gamma \subset Y$ satisfying

$$N\Gamma \leq (\dim Y + 1) \frac{NC}{-K_Y C}.$$

This statement is valid in arbitrary characteristic. Choose t so large that the ratio in Equation (3.8) is less than $1/4$. Applied on W with $N = D'$, the theorem gives, through every closed point $x \in C_t$, a rational curve Γ with

$$0 \leq D'\Gamma < 1.$$

As D' is Cartier, this degree is an integer and hence is zero.

We choose C_t to witness the very-general-point contradiction as well. Fix the product of the two chosen very ample linear systems, and let $\mathcal{T}$ be its nonempty open locus of smooth integral complete intersections avoiding the bad locus above. This parameter space is irreducible. Its universal curve $\mathcal{C} \rightarrow \mathcal{T}$ is smooth: each fiber is a transverse complete intersection inside W_{sm} , so the relative Jacobian criterion applies. Its evaluation map $\mathcal{C} \rightarrow W$ is dominant. Indeed the ambient incidence of points and pairs of divisors through them is a

product of projective bundles over W , hence is irreducible and surjects onto W ; restricting it to the nonempty open parameter locus $\mathcal{T}$ leaves a dense open subset, whose evaluation still has dense image.

Let $Z_1, Z_2, \dots$ be proper closed subsets of V outside whose union no D -trivial curve passes. The finite map π has proper closed images $\pi(Z_j)$. For each j , the locus in $\mathcal{T}$ of curves not contained in $\pi(Z_j)$ is the image of $\mathcal{C} \setminus \text{ev}^{-1}(\pi(Z_j))$ under the smooth map $\mathcal{C} \rightarrow \mathcal{T}$. It is open because smooth maps are open, and nonempty because evaluation is dominant. Over the uncountable field, the countably many resulting nonempty opens in $\mathcal{T}$ have a common k -point. On its integral curve C_t , each $C_t \cap \pi(Z_j)$ is finite. Choose a closed point x outside their union. Its unique inverse-image point $y \in V$ lies outside every Z_j .

The reduced inverse image of the rational curve Γ through x contains a curve $\tilde{\Gamma}$ through y , finite and surjective over Γ . Projection formula gives

$$pD\tilde{\Gamma} = (\pi^*D')\tilde{\Gamma} = \deg(\tilde{\Gamma}/\Gamma)D'\Gamma = 0.$$

This contradicts the choice of y . Thus T_V is semistable.

Suppose next that $e \geq 1$ is the first integer for which $F_V^{e*}T_V$ is unstable. Let A' be its first Harder–Narasimhan term and $Q' = F_V^{e*}T_V/A'$. Applying Lemma 3.1 to the semistable sheaf $F_V^{(e-1)*}T_V$ gives a nonzero map

$$A' \longrightarrow Q' \otimes \Omega_V^1, \quad \text{equivalently} \quad A' \otimes (Q')^\vee \longrightarrow \Omega_V^1$$

on a big open set. The cotangent bundle is semistable of slope zero, because T_V is. Since $\text{rk } A' + \text{rk } Q' = 3$, at least one tensor factor in the second source has rank one. If $\text{rk } Q' = 1$, semistability of A' and the rank-one twisting rule give the minimum slope of the source as $\mu(A') - \mu(Q')$. If $\text{rk } A' = 1$, the quotient Q' need not be semistable; use instead

$$\mu_{\min}((Q')^\vee) = -\mu_{\max}(Q')$$

and twist by the rank-one factor A' . In both cases,

$$\mu_{\min}(A' \otimes (Q')^\vee) = \mu(A') - \mu_{\max}(Q') > 0.$$

This contradicts Equation (3.2). All Frobenius pullbacks of T_V are therefore semistable.

Finally apply Langer's Bogomolov inequality [23, Theorem 3.2]; see also the addendum [22]. This theorem is formulated for a nef tuple $D_1, \dots, D_{n-1}$ with numerically nonzero product, and for a strongly semistable torsion-free sheaf gives $\Delta(E)D_2 \cdots D_{n-1} \geq 0$, where $\Delta(E) = 2\text{rk}(E)c_2(E) - (\text{rk}(E) - 1)c_1(E)^2$. Here the tuple is (H, D) , whose product is nonzero since $H^2D > 0$. For the rank-three tangent bundle this yields

$$(6c_2(V) - 2K_V^2)D \geq 0.$$

Equation (3.1) gives $K_V^2D = 0$ and proves Equation (3.7). $\square$

3.5 Uniform section bounds

Lemma 3.5. *If a line bundle L has $\deg L \leq 0$, then $h^0(V, L) \leq 1$.*

Proof. A nonzero section has an effective zero divisor, so its degree is nonnegative and hence must be zero. Suppose that two independent sections s_0, s_1 exist. Remove the base locus of the resulting pencil. Through every point x of the remaining open set there is a member of the pencil: take a nonzero linear combination vanishing at x . Its divisor has degree zero. Each prime component S through x therefore satisfies $DHS = 0$, because all component degrees are nonnegative.

The restriction of D to S is nef and numerically trivial. One way to see the latter assertion, allowing S to be singular, is to take a projective resolution $\sigma: \tilde{S} \rightarrow S$. The class $L_S = \sigma^*(D|_S)$ is nef and is orthogonal to the big and nef class $A_S = \sigma^*(H|_S)$, with $A_S^2 > 0$. The surface Hodge index theorem [15, Chapter V, Section 1] makes the intersection form negative definite on $A_S^\perp$, whereas nefness gives $L_S^2 \geq 0$. Hence L_S is numerically zero, and the projection formula gives the assertion on S . A projective integral curve in S through x , obtained for example as a component of a sufficiently ample section through x , consequently has D -degree zero. Choosing x very general contradicts the prepared-cycle property. $\square$

Lemma 3.6 (A uniform Hom bound). *If G_1, G_2 are torsion-free semistable sheaves of slope zero, then*

$$\dim_k \operatorname{Hom}(G_1, G_2) \leq \operatorname{rk}(G_1) \operatorname{rk}(G_2). \quad (3.9)$$

Proof. First suppose both sheaves are stable. For a nonzero map, its image has slope at least zero as a quotient of G_1 , and at most zero as a subsheaf of G_2 . A positive-rank kernel would make the first inequality strict by stability of G_1 . An image of rank smaller than $\operatorname{rk} G_2$ would make the second inequality strict by stability of G_2 . Thus every nonzero map has full rank, and the two sheaves have the same rank, say n .

The determinant of each map is a global section of

$$L = \det(G_2) \otimes \det(G_1)^{-1}, \quad \deg L = 0.$$

It is initially defined on the locally free open set and extends across its codimension-at-least-two complement. By Lemma 3.5, $H^0(V, L)$ has dimension at most one. If f, g were linearly independent maps, fix a nonzero section τ spanning this space and write

$$\det(sf + tg) = P(s, t)\tau$$

for a homogeneous polynomial P of degree n . Since f has full rank, P is nonzero; since $n > 0$ and k is algebraically closed, P vanishes at some point of $\mathbb{P}^1(k)$. The corresponding linear combination is a nonzero map by independence, but has zero determinant, a contradiction. Thus the Hom space between stable factors has dimension at most one.

Every slope-zero semistable sheaf admits a finite filtration by saturated subsheaves with torsion-free stable factors of slope zero. Indeed if it is not stable, a smaller-rank equal-slope subsheaf can be saturated; its degree cannot increase strictly because of semistability. Both it and the quotient are semistable of slope zero, and induction on rank finishes the construction. Apply the left-exact Hom sequences successively to such filtrations of G_1 and G_2 . The dimension of the Hom space is at most the sum of the dimensions between all pairs of stable factors. Each summand is at most one, and the numbers of factors are at most the respective ranks. This proves Equation (3.9). $\square$

Proposition 3.7 (Polar-part bounds). *Let G be a semistable vector bundle of slope zero. For every integer i ,*

$$h^0(V, G(iD)) \leq \operatorname{rk} G, \quad h^0(V, G^{\otimes 2}(iD)) \leq (\operatorname{rk} G)^2. \quad (3.10)$$

For each fixed bundle $J = G$ or $J = G^{\otimes 2}$, the increasing spaces

$$H^0(V, J(iD)/J), \quad i = 1, 2, \dots, \quad (3.11)$$

have bounded dimension and eventually stabilize under their natural injections. In particular their elements have a uniform bound on their actual pole orders along every prime component of D . This pole bound may depend on J ; it is uniform in the cut i and in the chosen section.

Proof. All the twists $G(iD)$ are semistable of slope zero. Apply Lemma 3.6 first to $\mathcal{O}_V, G(iD)$ and then to $G^\vee, G(iD)$. The identities

$$H^0(V, G(iD)) = \text{Hom}(\mathcal{O}_V, G(iD)), \quad H^0(V, G^{\otimes 2}(iD)) = \text{Hom}(G^\vee, G(iD))$$

give Equation (3.10). In particular the second bound uses the two semistable bundles $G^\vee$ and $G(iD)$, without needing semistability of $G^{\otimes 2}$.

For $i > 0$, the cohomology sequence of $0 \rightarrow J \rightarrow J(iD) \rightarrow J(iD)/J \rightarrow 0$ gives

$$h^0(V, J(iD)/J) \leq h^0(V, J(iD)) + h^1(V, J).$$

The right side is bounded independently of i . Effectivity of D gives injections of the quotient sheaves as i increases, hence injections of their finite-dimensional spaces of global sections. A bounded increasing sequence of integer dimensions is eventually constant, so these spaces stabilize.

More explicitly, choose i_0 after stabilization. Every global polar part, at every later cut, is then the image of an element of $H^0(V, J(i_0 D)/J)$. At the generic point of a prime component D_α of multiplicity m_α , a local bundle frame measures its pole order coefficientwise. Such an image has pole order at most $i_0 m_\alpha$ there. This bound is independent of the later cut and of the element chosen. $\square$

Corollary 3.8 (No scalar polar parts). *For every $i > 0$,*

$$H^0(V, \mathcal{O}_V(iD)/\mathcal{O}_V) = 0.$$

Proof. Apply Proposition 3.7 to $G = \mathcal{O}_V$. Suppose a nonzero scalar polar part η exists. It has a genuine negative valuation at the generic point of some component of D . To justify this detection, take local rational-function representatives. If all their component valuations were nonnegative, they would be regular by normality of V ; they already have no poles off D . The quotient section would then be zero.

For every $e \geq 0$, raising the local representatives to their p^e -th powers defines a global section

$$\eta^{p^e} \in H^0(V, \mathcal{O}_V(p^e iD)/\mathcal{O}_V).$$

This operation is well defined on polar parts: two representatives differing by a regular function have p^e -th powers differing by a regular function. At the selected component the negative valuation is multiplied by p^e . These pole orders are unbounded, contradicting the uniform bound in Proposition 3.7. $\square$

4 Jets and the three rows

We retain the prepared data of Proposition 2.1. Thus V is a smooth projective threefold, and

$$D = \sum_{\alpha} m_{\alpha} D_{\alpha}$$

is an effective Cartier divisor with connected support, $\gcd_{\alpha} m_{\alpha} = 1$, and $D|_{D_{\alpha}}$ numerically trivial for every α . In particular, $D^2 A = 0$ for every divisor A . The line bundle $\mathcal{O}_D(D)$ is torsion. Near D there is an isomorphism

$$\omega_V^{\otimes p^b} \simeq \mathcal{O}_V(B_*), \quad B_* = \sum_{\alpha} b_{\alpha} D_{\alpha}, \quad b_{\alpha} \in \mathbb{Z}, \quad (4.1)$$

and $K_V^2 D = 0$. We shall use $c_2(V)D \geq 0$ from Proposition 3.4 and the absence of scalar polar parts from Corollary 3.8.

The argument in this section concerns finite jets and their ranks. Its two outputs are a trivialization of the canonical line on the completion up to an integral divisor supported on D , and the equality $c_2(V)D = 0$ together with the limit $p/(p+1)$ for the normalized last trivialization widths. The width at period rp^e is the largest integer s for which $\mathcal{O}(rp^e D)$ is trivial on sD . Its only passage to a formal limit will be an explicit inverse-limit argument for finite-dimensional spaces of sections.

4.1 Finite pages and duality

Write $\widehat{\mathcal{O}}_V$ for the completion along D , regarded as a sheaf on $|D|$, and set

$$\mathcal{R} = \widehat{\mathcal{O}}_V(*D).$$

Choose a finite cover by the restrictions to $|D|$ of affine open subsets of V , refining it so that D is principal and the bundles under consideration are free on each member. All intersections of these affine opens are affine, since V is separated. For a vector bundle E , let $C_i^\bullet(E)$ be the resulting Čech complex with coefficients in the completion of $E(iD)$. These complexes give an increasing filtration, indexed by $i \in \mathbb{Z}$, on the Laurent Čech complex.

Here are the elementary completeness facts we need. On an affine chart with local equation f for D , adic completion is flat and f remains a non-zero-divisor. Thus the completed lattices embed in their localization. A section of the localization on a quasi-compact open lies in one lattice: finitely many local pole bounds have a common upper bound, and the resulting lattice sections glue. Consequently every Laurent cochain has a finite upper weight. Each $C_i^\bullet(E)$ is complete and separated for the filtration by $C_{i-s}^\bullet(E)$, $s \geq 1$. Exactness of completion for finite modules gives

$$\begin{aligned} C_i^\bullet(E)/C_{i-s}^\bullet(E) & \text{ computes } H^\bullet(V, P_i^s(E)), \\ P_i^s(E) &= \frac{E(iD)}{E((i-s)D)}. \end{aligned} \tag{4.2}$$

Indeed its terms are exactly the sections of this algebraic quotient on the affine intersections. Thus $P_i^s(E)$ retains the s layers at weights $i-s+1, \dots, i$; we also call i the upper cut. We abbreviate $E^\dagger = E^\vee \otimes \omega_V$.

For $s \geq 1$, consider a leading cochain, that is, a class in $C_i^a(E)/C_{i-1}^a(E)$, where a is its cohomological degree. The cycle condition below asks whether this class has a representative closed modulo $C_{i-s}^\bullet(E)$. The boundary condition asks whether it is the leading term of a coboundary with a primitive in $C_{i+s-1}^{a-1}(E)$. Thus the index s records both the precision of closedness and the allowed upper cut of a primitive. Define

$$\begin{aligned} Z_s^{i,a} &= \{[x] : x \in C_i^a(E), \partial x \in C_{i-s}^{a+1}(E)\}, \\ U_s^{i,a} &= \{[\partial y] : y \in C_{i+s-1}^{a-1}(E), \partial y \in C_i^a(E)\}. \end{aligned}$$

Every exact cochain is closed, so $U_s^{i,a} \subseteq Z_s^{i,a}$. Put

$$E_s^{i,a}(E) = Z_s^{i,a}/U_s^{i,a}, \quad g_s^a(i; E) = \dim_k E_s^{i,a}(E).$$

The notation $E_s^{i,a}(E)$ denotes a page space, not a new bundle.

Lemma 4.1 (Finite-page calculus). *The operation $x \mapsto \partial x$ induces maps*

$$d_s : E_s^{i,a}(E) \longrightarrow E_s^{i-s,a+1}(E)$$

whose cohomology is the next page. At page one, $E_1^{i,a}(E) = H^a(V, P_i^1(E))$; every page has only rows $a = 0, 1, 2$. In particular, with $\delta_s^a(i) = g_s^a(i; E) - g_{s+1}^a(i; E)$ for $a = 0, 2$,

$$g_s^0(i; E) = \dim \operatorname{im}(H^0(P_i^s(E)) \longrightarrow H^0(P_i^1(E))), \quad (4.3)$$

$$g_s^2(i; E) = g_s^0(1 - i; E^!), \quad (4.4)$$

$$g_{s+1}^1(i; E) = g_s^1(i; E) - \delta_s^0(i + s) - \delta_s^2(i - s). \quad (4.5)$$

If $M > 0$ and $\mathcal{O}(MD)|_{sD}$ is trivial, the page- s ranks are periodic in i modulo M , and

$$\sum_{i \bmod M} (g_s^0(i; E) - g_s^1(i; E) + g_s^2(i; E)) = \sum_{i \bmod M} \chi(P_i^1(E)). \quad (4.6)$$

Finally, at each fixed weight i , the stable image in Equation 4.3 is precisely the image of the global sections of the completed bundle $\widehat{E}(iD)$ in $H^0(P_i^1(E))$.

Proof. If two representatives of a leading cochain differ by $y \in C_{i-1}^a$, and both have coboundary in C_{i-s} , their coboundaries differ by a target boundary coming from $C_{i-1} = C_{(i-s)+s-1}$. Adding an exact representative has no effect either. This proves that d_s is well defined, and $d_s^2 = 0$ follows from $\partial^2 = 0$.

If $d_s[x] = 0$, the leading term of ∂x is represented by ∂y with $y \in C_{i-1}^a$ and $\partial y \in C_{i-s}^{a+1}$. Replacing x by $x - y$ makes its coboundary lie in C_{i-s-1}^{a+1} . Thus the kernel is represented by $Z_{s+1}^{i,a}$. The incoming image enlarges $U_s^{i,a}$ to $U_{s+1}^{i,a}$: its representatives are exactly the leading terms of $\partial z \in C_i^a$ with $z \in C_{i+s}^{a-1}$. This proves the next-page assertion. Equation 4.2 identifies page one. Since the quotients are supported on the projective surface scheme D , their cohomology is zero outside degrees 0, 1, 2; the same is true of every subsequent page.

The differentials through a middle-row term are arranged as

$$E_s^{i+s,0}(E) \xrightarrow{d_s} E_s^{i,1}(E) \xrightarrow{d_s} E_s^{i-s,2}(E).$$

Moving right lowers weight by s and raises cohomological degree by one; the composite is zero. There is no incoming differential in degree zero. Its representatives are exactly the sections of $P_i^s(E)$, proving Equation 4.3.

For the other endpoint, consider the inclusion of the bottom layer

$$H^2(P_i^1(E)) \longrightarrow H^2(P_{i+s-1}^s(E)).$$

There is no outgoing differential in degree two. More explicitly, the kernel calculation above and vanishing of the degree-three page imply successively that $Z_s^{i,2} = Z_1^{i,2}$. A class represented by $x \in C_i^2$, with $\partial x \in C_{i-1}^3$, dies under this inclusion exactly when $x - \partial y \in C_{i-1}^2$ for some $y \in C_{i+s-1}^1$. Its kernel is therefore $U_s^{i,2}/U_1^{i,2}$, and the image has dimension $g_s^2(i; E)$.

The two-term locally free presentation of $P_j^s(E)$ gives

$$\mathcal{E}xt^1(P_j^s(E), \omega_V) = \frac{E^!((s-j)D)}{E^!(-jD)} = P_{s-j}^s(E^!),$$

and all other sheaf Ext groups vanish. Serre duality on the smooth projective threefold consequently identifies $H^2(P_j^s(E))^\vee$ with $H^0(P_{s-j}^s(E^!))$ [15, Chapter III, Section 7]. Under this identification the dual of the displayed inclusion is

$$H^0(P_{1-i}^s(E^!)) \longrightarrow H^0(P_{1-i}^1(E^!)).$$

One can check the map, as well as its index, directly: the map of the Cartier presentations is the identity on $E((i-1)D)$ and the inclusion on their upper terms, so dualizing gives precisely

the indicated projection. This proves Equation 4.4. The ranks of the two differentials adjacent to the middle term are $\delta_s^0(i+s)$ and $\delta_s^2(i-s)$, respectively. Taking its cohomology proves Equation 4.5.

Triviality of $\mathcal{O}(MD)$ on sD identifies each quotient of width $t \leq s$ with its translate by M , compatibly with its layer maps. Equations 4.3 and 4.4 therefore give periodicity of both outer rows at every such width. Page one is periodic in all degrees; Equation 4.5 then gives periodicity of the middle row by induction. If $e_t(i) = g_t^0(i) - g_t^1(i) + g_t^2(i)$, that equation yields

$$e_{t+1}(i) - e_t(i) = -\delta_t^0(i) - \delta_t^2(i) + \delta_t^0(i+t) + \delta_t^2(i-t).$$

For $t < s$ all four functions are periodic modulo M . Their sums cancel, proving Equation 4.6 by induction from page one.

For the last assertion, set $A_s = H^0(P_i^s(E))$ and

$$B_s = \bigcap_{t \geq s} \text{im}(A_t \longrightarrow A_s).$$

Each A_s is finite dimensional, so the intersection defining B_s is attained at a finite stage. Choosing one stage beyond stabilization at both s and $s+1$ shows that $B_{s+1} \longrightarrow B_s$ is surjective. Every element of B_1 therefore admits compatible lifts in all the B_s . Their inverse limit is a global section of $\widehat{E(iD)}$, since sections commute with inverse limits of sheaves. Conversely a completed section restricts to the image of every A_s . This proves the assertion directly. $\square$

4.2 Scalar jets and their last widths

Lemma 4.2 (The scalar jet rule). *Let $u < i$ be integers and let $\xi \in H^0(\mathcal{O}(iD)/\mathcal{O}(uD))$ be nonzero. There is an integer t with $u < t \leq i$ such that local lifts of ξ have pole order exactly tm_α at every D_α . Those lifts give a trivializing section of $\mathcal{O}(tD)$ on $(t-u)D$. In particular,*

$$g_s^0(i; \mathcal{O}_V) = \begin{cases} 1, & \mathcal{O}(iD)|_{sD} \simeq \mathcal{O}_{sD}, \\ 0, & \text{otherwise.} \end{cases} \quad (4.7)$$

Proof. Choose local algebraic lifts f_λ of ξ on a finite affine cover. Membership in $\mathcal{O}(uD)$ is checked by valuations at the prime divisors of the smooth, hence normal, variety V . Thus some component detects the nonzero quotient section. The maximum of the ratios $-\text{ord}_{D_\alpha}(f_\lambda)/m_\alpha$ that exceed u is a well-defined rational number $t \in (u, i]$: changing a lift adds a section of $\mathcal{O}(uD)$ and cannot alter such a valuation. All lifts belong to $\mathcal{O}(tD)$ in the sense of these rational divisor bounds.

Choose a positive integer n with $nt \in \mathbb{Z}$. Then $f_\lambda^n \in \mathcal{O}(ntD)$, and these sections agree on D_{red} . Indeed $f_\lambda - f_\mu \in \mathcal{O}(uD)$, and in the factorization of $f_\lambda^n - f_\mu^n$ each term has one factor of pole ratio at most u and $n-1$ factors of pole ratio at most t . After normalization in $\mathcal{O}(ntD)$ its valuation at every D_α is at least $(t-u)m_\alpha > 0$. They therefore define a section of $\mathcal{O}(ntD)|_{D_{\text{red}}}$, nonzero on a component where the maximum is attained.

The restriction of this line bundle to every integral component is numerically trivial. A nonzero section of a numerically trivial line bundle on a projective integral surface has no zeros: a nonempty effective Cartier zero divisor has positive intersection with an ample divisor. It is consequently nowhere zero on that component. At an intersection with another component its common value is nonzero, so its restriction to that component is also nonzero and nowhere zero. Connectedness of $|D|$ propagates this conclusion to all components.

It follows that $\text{ord}_{D_\alpha}(f_\lambda) = -tm_\alpha$ for every α . In particular every tm_α is an integer. Primitivity of the cycle, or an integral linear combination of the m_α equal to one, gives $t \in \mathbb{Z}$.

The original lifts are now sections of the honest line bundle $\mathcal{O}(tD)$, are units in it along D , and agree modulo $\mathcal{O}(uD)$. Since

$$\frac{\mathcal{O}(tD)}{\mathcal{O}(uD)} = \mathcal{O}(tD)|_{(t-u)D},$$

they give the asserted trivialization.

A jet of width s at upper weight i has nonzero image in the leading layer exactly when the uniform ratio just proved is $t = i$. This proves existence in Equation 4.7, in both directions. To see that the image dimension is one, take two such jets. Their normalized unit sections on D_{red} are proportional: a connected reduced projective scheme over k has only constant global functions. Subtract the appropriate multiple. If its leading image were still nonzero, the preceding argument would again give a unit on the reduction with ratio i , contradicting the cancellation. Thus the two leading images are proportional. $\square$

The organizing viewpoint of tracking torsion and triviality on infinitesimal thickenings was informed by Totaro's treatment of these obstructions [29, Lemma 4.1 and Section 6]. The elementary width properties are proved below; the later three-row rank calculation and Cartier descent are not inputs from that work.

Let r be the order of $\mathcal{O}_D(D)$. For $e \geq 0$ put

$$M_e = rp^e, \quad d_e = \max\{s \geq 1 : \mathcal{O}(M_e D)|_{sD} \simeq \mathcal{O}_{sD}\}. \quad (4.8)$$

These maxima exist and satisfy

$$1 \leq d_e < M_e, \quad d_{e+1} \geq pd_e, \quad \frac{d_e}{M_e} \nearrow a_* \in (0, 1]. \quad (4.9)$$

In fact a trivialization on $M_e D$ would give, by Lemma 4.2, a nonzero section of $\mathcal{O}(M_e D)/\mathcal{O}$, contrary to Corollary 3.8. Triviality at width one follows from the definition of r . Taking p th powers of local trivializing sections multiplies the gluing precision by p and gives $d_{e+1} \geq pd_e$. Division by $M_{e+1} = pM_e$ proves monotonicity and the positive limit. If $r = 1$, already $1 \leq d_0 < M_0 = 1$ is a contradiction; we may therefore continue with $r > 1$.

We will repeatedly use the following precise consequence of nilpotence. If a line bundle L on a finite thickening T of D is trivial on D , then its class has p -power order. Indeed its transition functions can be chosen in $1 + J$, where J is the nilpotent ideal of D in T . For p^N at least a nilpotence exponent, $(1 + x)p^N = 1$ for $x \in J$. Thus $L^{\otimes p^N}$ is trivial. Multiplication by an integer prime to p is injective on such classes, by Bézout's identity, as well as surjective on their cyclic subgroups. It follows that

$$\text{the last width at } i = jM_e \text{ is } d_e \text{ if } p \nmid j. \quad (4.10)$$

This includes negative j . If $r \nmid i$, no scalar leading jet occurs even at width one. Every nonzero multiple of r has a unique expression jM_e with $p \nmid j$; at $i = 0$ the scalar entry persists at every width. These observations classify all scalar outer-row entries and drops.

Finally Hirzebruch–Riemann–Roch on the smooth threefold gives

$$\chi(P_i^1(\mathcal{O}_V)) = \chi(\mathcal{O}(iD)) - \chi(\mathcal{O}((i-1)D)) = \frac{Dc_2(V)}{12} =: \gamma \geq 0. \quad (4.11)$$

Indeed the terms involving D^3 , $D^2 K_V$, and DK_V^2 vanish, leaving the displayed linear term in the threefold Riemann–Roch polynomial [9, Section 15].

4.3 Canonical jets and a formal canonical root

For $E = \mathcal{O}_V$, Equation 4.4 computes the top-row rank at weight i from leading jets of ω_V at weight $1 - i$. We need to describe that row as explicitly as the scalar bottom row. We first show that canonical jets survive to arbitrarily large widths. Their p^b th powers then let the scalar rule identify $\hat{\omega}_V$ with the line of an integral divisor supported on D .

Lemma 4.3. *There are nonzero leading jets of ω_V of arbitrarily large width, allowing their upper weight to vary.*

Proof. Suppose instead that $g_s^0(l; \omega_V) = 0$ for every l and every $s \geq S$, for some S . By Equation 4.4, the top row for $E = \mathcal{O}_V$ then vanishes at all these widths. Take e sufficiently large that $s_e = d_{e-1} + 1 \geq S$. At page s_e the bottom row is one precisely at multiples of M_e , by Equation 4.10 and $s_e \leq d_e$. This page is periodic modulo M_e . Equations 4.6 and 4.11 give

$$\sum_{i \bmod M_e} g_{s_e}^1(i; \mathcal{O}_V) = 1 - M_e \gamma \leq 1.$$

At differential length d_e , the bottom entries at jM_e , $p \nmid j$, each drop by one. By Equation 4.5 these drops consume one middle-row dimension at $-d_e + jM_e$. Middle-row dimensions can only decrease as the page increases. Periodicity and the preceding upper bound therefore force the initial middle row to consist exactly of one entry at $-d_e \pmod{M_e}$.

Between pages s_e and $s_{e+1} = d_e + 1$ these are the only drops: there is no top row, and all scalar entries with a different last-width index either disappeared earlier or have last width at least d_{e+1} . Modulo $M_{e+1} = pM_e$, the drops remove all the p lifts of the middle entry except its unshifted lift. Thus at page s_{e+1} the middle row consists of one entry at $-d_e \pmod{M_{e+1}}$. Applying the same argument with $e + 1$ says that it must instead lie at $-d_{e+1} \pmod{M_{e+1}}$. But $0 < d_e < d_{e+1} < M_{e+1}$, so these residues are different. This contradiction proves the lemma. $\square$

Proposition 4.4 (The canonical formal root). *Every coefficient of B_* in Equation 4.1 is divisible by p^b . Writing $B_* = p^b B_0$ with B_0 integral, there is an isomorphism on the completion along D ,*

$$\hat{\omega}_V \simeq \hat{\mathcal{O}}_V(B_0). \quad (4.12)$$

Proof. Put $P = p^b$, and choose a sequence of nonzero leading canonical jets of widths $s \rightarrow \infty$, with upper weights l that may vary. Raising local representatives to the P th power and using Equation 4.1 gives a scalar jet with bounds

$$A = PlD + B_*, \quad B = P(l - s)D + B_*.$$

The lower bound is multiplied by P exactly because P is a power of the characteristic. Set $\beta_\alpha = b_\alpha/m_\alpha$ and choose an integer $c_* \geq \max_\alpha \beta_\alpha$. Enlarging both bounds gives a section of

$$\frac{\mathcal{O}((Pl + c_*)D)}{\mathcal{O}((P(l - s) + c_*)D)}.$$

This section is nonzero for all sufficiently large s , uniformly in l . To check this, at a component detecting the original leading layer, a local coefficient f in an ordinary frame of ω_V has $-\text{ord}_{D_\alpha}(f)/m_\alpha > l - 1$. Its powered scalar coefficient has pole ratio

$$-P \frac{\text{ord}_{D_\alpha}(f)}{m_\alpha} + \beta_\alpha > P(l - 1) + \min_\alpha \beta_\alpha.$$

The new lower bound is $P(l - s) + c_*$, which is smaller for large s .

Lemma 4.2 now shows that all these powered coefficients have one integral pole ratio t . The preceding lower estimate and the upper bound give a constant C , independent of s, l , such that

$$|t - Pl| \leq C, \quad (4.13)$$

$\mathcal{O}(tD)$ is trivial to a width at least $Ps - C$.

For each fixed J , widths greater than d_{J-1} can occur only at multiples of M_J , by the scalar classification; zero is allowed as a multiple. Since the widths in Equation 4.13 tend to infinity, we can assign integers $J \rightarrow \infty$ along this sequence such that $M_J \mid t$.

Fix a component D_α . If f is the original local coefficient in a frame of ω_V , the local generator of $\mathcal{O}(B_*)$ has order $-b_\alpha$. Consequently

$$b_\alpha - tm_\alpha = P \operatorname{ord}_{D_\alpha}(f). \quad (4.14)$$

For large J , the integer t is divisible by P . Equation 4.14 therefore shows that $P \mid b_\alpha$ for every α . This proves $B_* = PB_0$ with B_0 integral.

For the same sequence put $h_0 = t/P$. The order identity says that the original local representatives are sections of

$$L_{h_0} = \omega_V(-B_0 + h_0D)$$

near D : their coefficient orders are exactly those required by this line bundle. They have no additional poles near D , since the original representatives were sections of $\omega_V(lD)$ and membership in these divisor lattices is checked on the normal variety in codimension one. Moreover $L_{h_0}^{\otimes P} \simeq \mathcal{O}(tD)$, and the P th powers of these sections are units along all of D by Lemma 4.2. The original sections are therefore themselves units along D . They agree modulo the original lower lattice $\omega_V((l-s)D)$. Hence they trivialize L_{h_0} on the effective thickness

$$T = (h_0 - l + s)D - B_0.$$

By Equation 4.13, $|h_0 - l|$ is bounded independently of the sequence. Thus T contains nD for integers $n \rightarrow \infty$. Also $M_J \mid t$ implies $M_{J-b} \mid h_0$ once $J \geq b$. The line bundle $\mathcal{O}(h_0D)$ is therefore trivial on $d_{J-b}D$, and $d_{J-b} \rightarrow \infty$ by Equation 4.9. Removing this twist shows that $L = \omega_V(-B_0)$ is trivial on arbitrarily large finite thickenings nD .

We finally choose these trivializations compatibly. Apply the stabilized-image argument in the proof of Lemma 4.1 to the spaces $H^0(nD, L|_{nD})$, also including restriction to D_{red} . The latter line bundle is trivial, and its section space is one dimensional because D_{red} is connected, reduced, and projective. The image from every sufficiently large thickening contains a unit, hence is that full one-dimensional space. The surjective maps between stabilized images therefore lift a chosen nonzero section on D_{red} compatibly to all thickenings. Each lift is a unit, as may be checked after reduction, and their inverse limit trivializes $\hat{L}$. This gives Equation 4.12. $\square$

4.4 The two endpoint residues and the limiting width

Write $B_0 = \sum_\alpha b_{0,\alpha} D_\alpha$, and now use h for the integer offset

$$h = - \left\lfloor \min_\alpha \frac{b_{0,\alpha}}{m_\alpha} \right\rfloor, \quad q = 1 - h. \quad (4.15)$$

The formal isomorphism of Proposition 4.4 identifies the finite canonical jet lattices with scalar lattices having upper and lower bounds $lD + B_0$ and $(l-s)D + B_0$. All these finite quotients are algebraic sheaves supported on finite thickenings of D . Indeed the localized

$\widehat{\mathcal{O}}_V$ -linear isomorphism respects every nested divisor lattice. Each width- s quotient is killed by $\mathcal{I}_D^s$, where $\mathcal{I}_D = \mathcal{O}_V(-D)$; since $\widehat{\mathcal{O}}_V/\mathcal{I}_D^s\widehat{\mathcal{O}}_V = \mathcal{O}_{sD}$, the induced map is an algebraic $\mathcal{O}_{sD}$ -linear map, compatible with the leading-layer projections.

We first give the exact canonical counterpart of the scalar rule. For all sufficiently large s , a canonical jet at weight l with nonzero leading image has uniform integral scalar pole ratio

$$t = l - h. \quad (4.16)$$

To prove this, enlarge the two bounds to uniform multiples of D as in the preceding proof. A component detecting the leading image has ratio within a fixed distance of l , so for sufficiently large s the enlarged quotient still detects it. Lemma 4.2 gives a uniform integral ratio t . Set $\beta_0 = \min_{\alpha} b_{0,\alpha}/m_{\alpha}$. Fitting under every upper bound, but not under the preceding layer, is exactly the pair of inequalities

$$l - 1 + \beta_0 < t \leq l + \beta_0.$$

Its unique integral solution is $t = l + \lfloor \beta_0 \rfloor = l - h$.

The normalized unit sections of $\mathcal{O}(tD)$ glue on the irregular thickness

$$T_s = (s - h)D - B_0, \quad (4.17)$$

because this is the difference between tD and the lower bound $(l - s)D + B_0$. Conversely a trivialization of $\mathcal{O}(tD)$ on T_s , with $t = l - h$, includes in the canonical quotient and has nonzero leading image. Indeed $tD \leq lD + B_0$, whereas the inequalities above prevent inclusion in the preceding upper lattice. The leading image has dimension one: normalized unit sections are proportional on D_{red} , and their difference cannot retain the same leading ratio after that reduction cancels, by the scalar rule. We have proved

$$g_s^0(l; \omega_V) = \begin{cases} 1, & \mathcal{O}((l - h)D)|_{T_s} \simeq \mathcal{O}_{T_s}, \\ 0, & \text{otherwise,} \end{cases} \quad (s \text{ sufficiently large}). \quad (4.18)$$

The comparison with ordinary widths is uniform and quite explicit. Put

$$C_0 = h + \left\lceil \max_{\alpha} \frac{b_{0,\alpha}}{m_{\alpha}} \right\rceil \geq 0.$$

Then, coefficient by coefficient,

$$(s - C_0)D \leq T_s \leq sD. \quad (4.19)$$

For e sufficiently large, let d'_e be the last width s for which $\mathcal{O}(M_e D)|_{T_s}$ is trivial. The thicknesses are nested as s increases. Equation 4.19 and the definition of d_e give

$$d_e \leq d'_e \leq d_e + C_0. \quad (4.20)$$

In particular this last width is finite and exists in the range where Equation 4.18 applies. Once T_s contains D , the nilpotent-kernel argument used above shows that the last width is the same for $t = jM_e$ whenever $p \nmid j$. Also such a trivialization requires $r \mid t$; for $t \neq 0$ the unique expression $t = jM_e$, $p \nmid j$, classifies its last width. The value $t = 0$ persists forever. To check that the large- e rules exhaust the entries at large width, observe from Equation 4.19 that an entry at width s gives an ordinary scalar trivialization to width $s - C_0$. For each fixed J , taking $s - C_0 > d_{J-1}$ forces $M_J \mid t$. Thus every nonzero index occurring at sufficiently large width belongs to the stated range of last-width indices.

Proposition 4.5 (The limiting scalar widths). *The scalar widths satisfy*

$$a_* = \lim_{e \rightarrow \infty} \frac{d_e}{M_e} = \frac{p}{p+1}, \quad Dc_2(V) = 0. \quad (4.21)$$

Proof. For all sufficiently large e set

$$s_e = \max(d_{e-1}, d'_{e-1}) + 1.$$

Equations 4.9 and 4.20 show that these are strictly increasing and $s_e \leq \min(d_e, d'_e)$. Work with the three rows for $E = \mathcal{O}_V$. At page s_e all three rows are periodic modulo M_e , and each outer row has a single entry per period. The bottom entry is at 0, by Equation 4.10. For the top entry, Equations 4.4 and 4.16 associate to a top weight i the scalar ratio $t = 1 - i - h = q - i$. Its entry is therefore at $q \pmod{M_e}$.

As the page runs from s_e through s_{e+1} , the only outer drops are those with last-width index e . The bottom drops, of length d_e , consume middle-row dimensions at

$$-d_e + jM_e, \quad p \nmid j.$$

The top drops, of length d'_e , consume middle-row dimensions at

$$q + d'_e + jM_e, \quad p \nmid j;$$

here replacing j by $-j$ accounts for the sign in the top ratio. Put

$$A_e = -d_e, \quad B_e = q + d'_e.$$

The two lists, including their multiplicities when they meet, are thus $A_e + jM_e$ and $B_e + jM_e$, with $p \nmid j$.

We spell out the rank count also when the two centers coincide modulo M_e . Fix a residue $a \pmod{M_e}$ and let $m(a) \in \{0, 1, 2\}$ be its multiplicity in the multiset $\{A_e, B_e\}$. Its initial middle-row rank $n(a)$ is constant on all p lifts modulo pM_e . Each of the $m(a)$ drop lists omits exactly one lift, namely its unshifted center modulo pM_e . At most two lifts are omitted.

For example, when $p = 5$ and $m(a) = 2$, two distinct omitted lifts can be pictured as follows:

	a	$a + M_e$	$a + 2M_e$	$a + 3M_e$	$a + 4M_e$
first list	○	●	●	●	●
second list	●	●	○	●	●

A filled dot consumes one rank; a circle marks that list's unshifted lift. The omitted lifts are illustrative. If they coincide, their column is omitted twice and every other column receives both drops. Multiplicities are always added.

In general, since $p > 2$, some lift receives all $m(a)$ drops. Nonnegativity of its final rank gives $n(a) \geq m(a)$. Summing over a , and then using Equations 4.6 and 4.11, gives

$$2 \leq \sum_{a \pmod{M_e}} n(a) = 2 - M_e \gamma \leq 2.$$

Consequently $\gamma = 0$, so $Dc_2(V) = 0$, and $n(a) = m(a)$ for every residue.

After subtracting the drop lists, the final rank at each lift is exactly the number of lists omitting that lift. Thus at page s_{e+1} the middle row is described by the same two-element multiset $\{A_e, B_e\}$, now modulo $M_{e+1} = pM_e$. Applying the just-proved description at index $e + 1$ gives the multiset equality

$$\{-d_e, q + d'_e\} = \{-d_{e+1}, q + d'_{e+1}\} \pmod{M_{e+1}}. \quad (4.22)$$

The entries $-d_e$ and $-d_{e+1}$ are distinct modulo M_{e+1} because $0 < d_e < d_{e+1} < M_{e+1}$. Therefore the matching in Equation 4.22 must be the swapped matching, even if a multiset initially has repeated entries. In particular

$$d_{e+1} + q + d'_e \equiv 0 \pmod{M_{e+1}}.$$

For large e its left side is positive and, by $d_{e+1} < M_{e+1}$ and $d'_e \leq d_e + C_0 < M_e + C_0$, is less than $2M_{e+1}$. Hence

$$d_{e+1} + q + d'_e = M_{e+1}. \quad (4.23)$$

Divide by $M_{e+1} = pM_e$ and let $e \rightarrow \infty$. Equation 4.20 gives $d'_e/M_{e+1} \rightarrow a_*/p$, while $d_{e+1}/M_{e+1} \rightarrow a_*$. Thus $a_* + a_*/p = 1$, which proves Equation 4.21. $\square$

5 Laurent cohomology of vector bundles

We retain the smooth threefold V , the cycle D , and the notation of the preceding sections. In particular,

$$M_e = rp^e, \quad \frac{d_e}{M_e} \rightarrow a_* := \frac{p}{p+1}, \quad c_2(V) \cdot D = 0. \quad (5.1)$$

The line bundle $\mathcal{O}_V(M_e D)$ is trivial through width d_e . Choose, once and for all, real numbers

$$1 - \frac{2}{p} < c_0 < c < 2a_* - 1. \quad (5.2)$$

Such choices exist since

$$2a_* - 1 - \left(1 - \frac{2}{p}\right) = \frac{2}{p(p+1)} > 0.$$

As $p \geq 5$, we also have $1/2 < c_0 < c < a_*$. Put $u_e = \lfloor cM_e \rfloor$. The lower bound on c_0 permits the use of only first and second tensor powers in Lemma 5.1; the upper bound on c leaves the precision required for matching jets in Theorem 5.2.

5.1 Concentration of leading jets

The first step restricts the positions at which a long jet of a semistable bundle can have a nonzero leading term.

Lemma 5.1. *Let G be a vector bundle that is semistable of slope zero for the degree $c_1(-) \cdot H \cdot D$. There is a constant A_G such that, for every sufficiently large e ,*

$$g_{\lfloor c_0 M_e \rfloor}^0(i; G) > 0 \implies \text{dist}(i, M_e \mathbb{Z}) \leq A_G.$$

Proof. Write $s = \lfloor c_0 M_e \rfloor$. For large e we have $s \leq d_e$. A unit of $\mathcal{O}(M_e D)|_{sD}$ translates the upper weight of a jet by M_e . We can therefore replace i by a centered representative v , with $|v| \leq M_e/2$, without changing the existence of its leading class.

Suppose that no uniform bound on $|v|$ exists. Passing to a sequence, we may assume that $v \rightarrow +\infty$ or $v \rightarrow -\infty$. In the first case $s > v$ for large e . The jet maps to a global section of $G(vD)/G$ and retains its nonzero leading term. Its pole order tends to infinity, contrary to Proposition 3.7.

In the second case choose a real number λ such that

$$\frac{1 - c_0}{2} < \lambda < \frac{1}{p}, \quad p\lambda > \frac{1}{2}. \quad (5.3)$$

The interval is nonempty by (5.2). Since the periods increase by the factor p , we can choose $j \leq e$ with

$$\lambda \leq \frac{|v|}{M_j} \leq p\lambda.$$

Here $j \rightarrow \infty$ as $|v| \rightarrow \infty$. After a subsequence the ratios tend to a number $y \in [\lambda, p\lambda]$. There is an integer $m \in \{1, 2\}$ for which

$$1 - c_0 < my < 1. \quad (5.4)$$

Indeed, if $y > 1 - c_0$, take $m = 1$; then $y \leq p\lambda < 1$. Otherwise take $m = 2$. Its lower bound follows from $2y \geq 2\lambda > 1 - c_0$, and its upper bound follows from $2y \leq 2(1 - c_0) < 4/p < 1$.

Restrict the original jet to width $\lfloor c_0 M_j \rfloor$, take its m th tensor power, and multiply by a unit section of $\mathcal{O}(M_j D)$ to that width. Tensoring keeps at least the same width: in a difference of two tensor products, every summand contains a factor from the lower lattice. The resulting jet is a jet of $G^{\otimes m}$ at upper weight

$$w = M_j + mv.$$

By (5.4), $w \rightarrow +\infty$ and $w < \lfloor c_0 M_j \rfloor$ with a margin tending to infinity. It therefore gives a global polar part of $G^{\otimes m}$.

This polar part has unbounded actual order. At some component of D an original coefficient has pole ratio greater than $v - 1$. The corresponding pure tensor coefficient, after multiplication by the unit, has pole ratio greater than $M_j + m(v - 1)$. This tends to infinity. Proposition 3.7 applies to both G and $G \otimes G$, giving the required contradiction. $\square$

5.2 A vanishing criterion

We use $\widehat{E}$ for the completion of a bundle E along D , and $E_{\mathcal{R}} = \widehat{E} \otimes_{\widehat{\mathcal{O}}} \mathcal{R}$. The finite-cover, bounded-pole interpretation of these sheaves is the one used in the jet construction.

Theorem 5.2. *For the prepared data and periods fixed above, let E be a vector bundle on V with the following properties:*

- (i) $\chi(P_i^1(E)) = 0$ for every integer i .
- (ii) For $J = E, E^!$, the spaces $H^0(V, J(nD)/J)$ have dimensions bounded independently of the positive integer n .
- (iii) For $J = E, E^!$ there is a constant A_J such that

$$g_{u_e}^0(i; J) > 0 \implies \text{dist}(i, M_e \mathbb{Z}) \leq A_J$$

for all sufficiently large e .

Then global sections of $E_{\mathcal{R}}$ form a finite-dimensional k -vector space. Every Laurent Čech 1-cocycle on a finite affine cover adapted to the bundles and the divisor has an actual Laurent primitive. The same assertion holds after refining such a cover.

Proof. We separate the argument into the endpoint patterns, their drops, the middle row, and convergence.

Endpoint patterns. For e large, $u_e \leq d_e$, so the page at width u_e is periodic modulo M_e . By (iii), the bottom row can be supported only at a fixed bounded set of centered positions. At each such position the descending sequence of leading images stabilizes as the width tends to infinity. The stable-image statement of Lemma 4.1 realizes every vector in the stabilized image by a global completed section at that cut. Consequently, for all sufficiently large e there is one fixed bottom pattern, at positions v with fixed multiplicities, repeated by translations

by M_e . Duality gives a fixed top pattern at positions w . We enlarge a constant A so that all these v, w lie in $[-A, A]$.

This already proves finite-dimensionality of global Laurent sections. A nonzero section has a least upper cut i : its poles are bounded, and the filtration is separated toward negative infinity. Its nonzero leading class belongs to $g_{u_e}^0(i; E)$ for every large e . With i fixed and $M_e \rightarrow \infty$, condition (iii) forces i to be an unshifted position of the fixed pattern. There are only finitely many such cuts, and each leading quotient is finite-dimensional. These quotients give a finite filtration of the space of global sections. The same observation applies to $E^!$.

Endpoint drops. By the middle-row recurrence, a bottom drop of length s at $v + jM_e$ removes middle-row rank at $v - s + jM_e$; a top drop at $w + jM_e$ removes rank at $w + s + jM_e$. To determine these losses over the next period, we must control their lengths, with multiplicities, independently of j whenever $p \nmid j$. Consider a bottom position $i = v + jM_e$. If $p \mid j$, the next period $M_{e+1} = pM_e$ retains exactly the same multiplicity there. Monotonicity therefore permits no drop between pages u_e and u_{e+1} . If $p \nmid j$, all of this multiplicity disappears by page u_{e+1} . We claim that its drops occur at lengths

$$d_e \leq s \leq M_e + C \quad (5.5)$$

for a fixed C , and that their multiplicities at each length are independent of j among integers prime to p .

There is no drop before length d_e , since translation by a unit of $\mathcal{O}(M_e D)|_{d_e D}$ identifies the leading images with those at the unshifted stable position v . Choose local formal unit lifts t_λ of this trivializing section of $\mathcal{O}(M_e D)|_{d_e D}$, and write on overlaps

$$\frac{t_\lambda}{t_\mu} = 1 + b_{\lambda\mu}, \quad b_{\lambda\mu} \in \mathcal{I}_D^{d_e} \widehat{\mathcal{O}}_V, \quad \mathcal{I}_D = \mathcal{O}_V(-D).$$

Fix a constant C for the moment and a width

$$d_e < k \leq M_e + C.$$

For large e , $k < 2d_e$ by (5.1). Normalize local representatives σ_λ of a width- k jet at $v + jM_e$ by putting $\alpha_\lambda = \sigma_\lambda / t_\lambda^j$. These are local sections of $\widehat{E(vD)}$ and satisfy

$$\alpha_\mu = (1 + jb_{\lambda\mu})\alpha_\lambda \pmod{\mathcal{I}_D^k \widehat{E(vD)}}. \quad (5.6)$$

Indeed $(1 + b_{\lambda\mu})^j = 1 + jb_{\lambda\mu}$ to this precision, for positive or negative j alike, because $k < 2d_e$. In particular, the cochain α glues to width d_e .

We next match α to a single global completed section β in that untwisted lattice through width $k - d_e$. Every completed section and every formal lattice map restricts to the actual algebraic finite jets of Equation (4.2): on each affine intersection this is the identity $\widehat{A}/(f^s) = A/(f^s)$. These restrictions commute with the Čech maps and with projection to leading layers. Start with $\beta_{<0} = 0$. After matching the first n depths, the sum $\beta_{<n}$ of the completed global sections already chosen satisfies

$$\alpha - \beta_{<n} \in C_{v-n}^0(E), \quad \partial(\alpha - \beta_{<n}) \in C_{v-d_e}^1(E).$$

The second inclusion persists because each subtracted section is global. Thus, for $n < k - d_e$, the residual is a finite jet at upper weight $v - n$ of width at least $d_e - n$. The inequalities

$$d_e - n \geq 2d_e - M_e - O(1) > u_e, \quad k - d_e < M_e/2 - O(1) \quad (5.7)$$

hold with margins growing linearly in M_e , by $c < 2a_* - 1$ and $a_* > 1/2$. Thus $v - n$ is centered, and condition (iii) says that its leading image is either zero or one of the unshifted

stabilized images. In the latter case lift it to a completed global section and subtract. A finite induction constructs the asserted β .

If $p \nmid j, j'$, put $q = j'/j$ in the prime field and set

$$\alpha' = q\alpha + (1 - q)\beta.$$

Since β is global, Equation (5.6) gives

$$\alpha'_\mu - (1 + j'b_{\lambda\mu})\alpha'_\lambda = j'(1 - q)b_{\lambda\mu}(\alpha_\lambda - \beta) \pmod{\mathcal{I}_D^k \widehat{E(vD)}}.$$

The right side vanishes: its two factors have depths d_e and $k - d_e$. Retwisting by $t_\lambda^{j'}$ therefore gives a width- k jet at $v + j'M_e$. It has the same normalized leading term, since $k - d_e \geq 1$. Also $\alpha' - \beta = q(\alpha - \beta)$, so the same β reverses the construction with q^{-1} . The normalized leading-image subspaces are consequently equal at every width under consideration. Their ranks, and hence their drops at each length, are independent of j prime to p .

For $j = 1$, property (ii) kills the entire leading image by width $M_e + C$ once $C > A$ is sufficiently large. Indeed this width is at least the upper weight $M_e + v$, so a surviving leading jet would give a global polar part of order tending to infinity. The increasing spaces in (ii) stabilize and forbid such polar parts. Since $u_{e+1} > M_e + C$ for large e , this proves (5.5) and the complete drop rule. The identical argument for $E^\dagger$, followed by the duality of Lemma 4.1, gives the corresponding top-row rule.

The middle row. We use the nonnegativity and Euler comparison from Proposition 4.5, now with a multiset of drop positions. Let T_e be the multiset of the numbers $v - s$ for the bottom drops and $w + s$ for the top drops. Each is counted with its drop multiplicity for one translating integer prime to p . Its total size is the sum of the ranks of the two fixed endpoint patterns. Put $H_e(i) = g_{u_e}^1(i; E)$. The middle-row recurrence in Lemma 4.1 gives

$$\begin{aligned} H_{e+1}(i) = H_e(i) - \#\{t \in T_e : t \equiv i \pmod{M_e}\} \\ + \#\{t \in T_e : t \equiv i \pmod{pM_e}\}. \end{aligned} \quad (5.8)$$

The last term restores the unshifted copies, which do not drop. All members of T_e lie in the two intervals

$$[-M_e - O(1), -d_e + O(1)] \quad \text{and} \quad [d_e - O(1), M_e + O(1)]. \quad (5.9)$$

Each interval has length less than M_e for large e . For any fixed residue modulo M_e , these intervals can occupy at most two of its lifts modulo pM_e . Since $p \geq 5$, some lift is missed. Nonnegativity of H_{e+1} at that lift and periodicity of H_e show that

$$H_e(i) \geq \#\{t \in T_e : t \equiv i \pmod{M_e}\}.$$

Summing over a period gives equality of the two sides, by the Euler identity in Lemma 4.1 and property (i). The inequality is therefore equality at every residue. Equation (5.8) now says that H_{e+1} is exactly the distribution of T_e modulo pM_e . For each fixed integer i , the bands (5.9) and $d_e \rightarrow \infty$ imply

$$g_{u_e}^1(i; E) = 0 \quad \text{for all sufficiently large } e. \quad (5.10)$$

Convergence of primitives. Fixed-weight page vanishing alone does not bound the poles of successive primitives. Fix an integer upper bound l for a Laurent cocycle. We will bound the

source $z = x + s$ of every nonzero incoming differential of length s to a middle-row weight $x \leq l$ by a function $B_l(x)$ satisfying

$$\sup_{x \leq l} B_l(x) < \infty, \quad B_l(x) \longrightarrow -\infty \quad (x \rightarrow -\infty). \quad (5.11)$$

At a sufficiently late period, the drop rule gives $z = v + jM_e$, $p \nmid j$, and $d_e \leq s \leq M_e + C$. If $j \geq 2$, then $x \geq M_e - A - C$, which exceeds l for all sufficiently large e . If $j = 1$, the target must lie in the fixed finite set of integers in $[-A - C, l]$. Choose one index e_0 , depending on l , beyond the thresholds for the endpoint and drop rules, so that $M_e - A - C > l$ for $e \geq e_0$ and every middle rank in this finite set is zero at page u_{e_0} . Equation (5.10) permits this simultaneous choice. Middle ranks only decrease, so no nonzero incoming image can have $j = 1$ at any later page.

Put $S_l = u_{e_0} \geq 1$. Earlier lengths $s < S_l$ give $z \leq x + S_l$. Every later length belongs to a period $u_e \leq s < u_{e+1}$ with $e \geq e_0$; the preceding exclusions leave only $j \leq -1$. In that case $z \leq -M_e + A$, and hence $M_e \leq -z + A$. Thus $x = z - s \geq 2z - A - C$, or

$$z \leq \frac{x + A + C}{2}.$$

The real-valued upper bound

$$B_l(x) = \max \left\{ x + S_l, \frac{x + A + C}{2} \right\} \quad (x \leq l)$$

therefore bounds every possible source and satisfies both assertions of Equation (5.11).

Let γ be an actual Laurent 1-cocycle of upper cut $x \leq l$. Its leading cochain belongs to every cycle space $Z_s^{x,1}$. By (5.10), it eventually belongs to the boundary space $U_s^{x,1}$. More precisely, the page construction identifies

$$U_{s+1}^{x,1}/U_s^{x,1} = \text{im}(d_s : E_s^{x+s,0} \longrightarrow E_s^{x,1}),$$

where $E_s^{i,a} = Z_s^{i,a}/U_s^{i,a}$. Thus $U_s^{x,1}$ stops increasing once $x + s > B_l(x)$. By its definition, each increase from a source cut $z = x + s$ has a representative primitive in $C_z^0(E)$; the initial space $U_1^{x,1}$ uses $C_x^0(E)$. Consequently the leading cochain of γ has a primitive of cut at most $\max\{x, B_l(x)\}$. Subtract that primitive's coboundary. The new cocycle has strictly smaller upper cut. Repeating at successive descending cuts gives primitives whose upper cuts tend to negative infinity by (5.11). Their sum converges in a single bounded-pole completed lattice. The differential is continuous, and the filtration is separated, so its coboundary is γ . All arguments use finite covers by affine charts and their affine intersections, and remain valid after refinement. This proves the theorem. $\square$

5.3 Cartier algebra on the Laurent charts

We specify the module structures before applying the criterion to exact forms. Write F for absolute Frobenius and $\Omega = \Omega_V^1$. Let $B^1, Z^1 \subset F_*\Omega$ be the sheaves of exact and closed algebraic one-forms, with the Frobenius module structure.

Lemma 5.3. *On a smooth affine chart with étale coordinates x_1, x_2, x_3 and a local equation f for D , put $R = \widehat{A}[1/f]$. The monomials $x^I = x_1^{I_1} x_2^{I_2} x_3^{I_3}$, $0 \leq I_j < p$, form a basis of R over R^p . Ordinary differentials and the Cartier isomorphism extend to this ring. The algebraic bundles B^1, Z^1 and their defining exact sequences extend to the corresponding Laurent exact and closed forms, with their Frobenius module structure. In particular,*

$$0 \longrightarrow \mathcal{O}_V \longrightarrow F_*\mathcal{O}_V \longrightarrow B^1 \longrightarrow 0, \quad 0 \longrightarrow B^1 \longrightarrow Z^1 \xrightarrow{C} \Omega \longrightarrow 0 \quad (5.12)$$

are sequences of vector bundles and remain exact on Laurent charts. The images and kernels in the Frobenius de Rham complex are locally free, as are the relevant quotients. Moreover,

$$(F_*\mathcal{O}_V)^\vee \otimes \omega_V \simeq F_*\omega_V. \quad (5.13)$$

Proof. The algebraic calculation is the p -basis form of Cartier's theorem [2, Theorem 1]; see also [19, Theorem 7.2]. We first check that its finite basis survives the completion and localization used here. For an étale algebra the relative Frobenius square is a base change square. One can check this because the relative Frobenius is étale and induces isomorphisms on geometric fibers. The polynomial-coordinate monomial basis over p th powers therefore gives the asserted basis for A over A^p . The field k is perfect.

Complete the p -power subring along f^p and use this finite basis. The f -adic and f^p -adic topologies on A coincide, and Frobenius identifies the completion of A^p with $(\widehat{A})^p$. The basis thus persists after completion and then after inverting f ; in the latter step, $1/f = f^{p-1}/f^p$.

Every k -derivation kills p th powers. Since each element has a unique finite expansion in the displayed basis with coefficients in R^p , its differential is determined by ordinary differentiation of those monomials. Conversely the coordinate derivations are defined by that rule. Hence $\Omega_{R/k}^1$ is free on dx_j . There is no additional continuity assumption on differentials.

In one variable, the de Rham complex decomposes into the monomials of degrees less than p ; its cohomology has representatives 1 in degree zero and $x^{p-1}dx$ in degree one. Tensoring the three one-variable complexes proves local freeness of images, kernels, and quotients, and proves the inverse Cartier rule

$$f \longmapsto f^p, \quad df \longmapsto [f^{p-1}df].$$

The rule is intrinsic. Leibniz holds directly, while additivity for differentials follows modulo exact forms by differentiating the mixed terms of the integral polynomial $((X+Y)^p - X^p - Y^p)/p$ and reducing modulo p . Thus the local isomorphisms glue.

More explicitly, the scalar extension of the algebraic Frobenius complex to a Laurent chart is the actual Laurent de Rham complex with Frobenius action, through

$$R \otimes_A F_*\Omega_A^\bullet \longrightarrow F_*\Omega_R^\bullet, \quad a \otimes \eta \longmapsto a^p\eta.$$

The monomial and coordinate-wedge bases make this an isomorphism. This proves the assertions about (5.12). Finally, the pairing obtained by Cartier on top forms is perfect: in coordinates it extracts the coefficient of $x_1^{p-1}x_2^{p-1}x_3^{p-1}dx_1 \wedge dx_2 \wedge dx_3$. It gives (5.13) and is intrinsic by the same Cartier rule. $\square$

5.4 Application to one-forms and exact forms

Corollary 5.4. *Every Laurent Čech 1-cocycle with values in Ω , B^1 , or Z^1 has a Laurent primitive. Global Laurent one-forms form a finite-dimensional k -vector space.*

Proof. For $E = \Omega$, both E and $E^\dagger = T_V \otimes \omega_V$ are semistable of slope zero by Proposition 3.4 and the canonical intersection identities. Proposition 3.7 and Lemma 5.1 give conditions (ii) and (iii) of Theorem 5.2; passing from width $\lfloor c_0 M_e \rfloor$ to u_e can only decrease the leading images. Hirzebruch–Riemann–Roch gives condition (i). Indeed the degree-two terms in $\mathrm{ch}(\Omega) \mathrm{td}(T_V)$ are linear combinations of K_V^2 and $c_2(V)$, both killed by intersection with D , and every term involving D^2 is also zero.

For $E = B^1$, the first sequence in (5.12) gives condition (i). By projection formula, a single cell for $F_*\mathcal{O}_V$ has the Euler characteristic of p successive scalar cells; each is zero by Proposition 4.5. Frobenius changes the k -module action but not these finite dimensions.

There are subbundle inclusions with locally free quotients

$$B^1 \hookrightarrow F_*\Omega, \quad (B^1)^\dagger \hookrightarrow F_*\omega_V. \quad (5.14)$$

The first follows from Lemma 5.3; the second follows by dualizing the first sequence in (5.12) and using (5.13). For a subbundle $E \subset F_*G$, the injection remains injective on every quotient $E(nD)/E$. Projection formula embeds that quotient in $F_*(G(pnD)/G)$. The polar bounds for $G = \Omega$ and $G = \omega_V$ therefore give condition (ii) for both bundles in (5.14). All these bounds concern fixed bundles and their first Frobenius pushforwards. The growing index e below changes the jet period, not the bundle in the polar-part estimate.

The same subbundle property gives condition (iii), with a small precision margin that is useful to make explicit. A width- u_e jet of $E \subset F_*G$ with a nonzero leading term maps to a width- pu_e jet of G at upper weight pi . Its leading term is nonzero in the top p ordinary layers. Choose its first nonzero ordinary layer, at a weight a with $|a - pi| < p$. It has width at least $pu_e - p$. Since $c > c_0$ and $M_{e+1} = pM_e$, for all sufficiently large e we have

$$pu_e - p \geq \lfloor c_0 M_{e+1} \rfloor.$$

Lemma 5.1 at period M_{e+1} implies that a , and hence pi , is at bounded distance from $M_{e+1}\mathbb{Z}$. Dividing by p gives the required bounded distance from $M_e\mathbb{Z}$.

Theorem 5.2 now applies to Ω and B^1 , including finite-dimensionality for the former. For a Z^1 cocycle, first solve its image in Ω and lift the resulting local primitives through Cartier. After subtraction the remaining cocycle lies in B^1 and can also be solved. These lifts exist on affine Laurent charts by Lemma 5.3; refining the finite cover if necessary preserves every preceding argument. This proves the assertion for Z^1 . $\square$

The next section uses both conclusions: cocycle vanishing supplies compatible closed connection forms, and finite-dimensionality permits a global adjustment that makes those forms Cartier-fixed.

6 Cartier descent and the final degree contradiction

We work on the Zariski topological space $|D|$, with

$$\hat{\mathcal{O}} = \hat{\mathcal{O}}_V, \quad \mathcal{R} = \hat{\mathcal{O}}_V(*D).$$

Here and throughout this section, a line bundle means a locally free module of rank one for the indicated sheaf of rings, in this Zariski topology. We write $\Omega_{\mathcal{R}}^j$ for the actual Laurent differential forms, and $\mathcal{B}^1 \subset \mathcal{Z}^1 \subset \Omega_{\mathcal{R}}^1$ for the additive sheaves of exact and closed one-forms. Lemma 5.3 identifies these with the Laurent extensions of the algebraic Frobenius modules. In particular, there is an exact sequence of additive sheaves

$$0 \longrightarrow \mathcal{B}^1 \longrightarrow \mathcal{Z}^1 \xrightarrow{C} \Omega_{\mathcal{R}}^1 \longrightarrow 0, \quad (6.1)$$

where C is inverse-Frobenius-semilinear for the ordinary scalar structures:

$$C(a^p\alpha) = aC(\alpha).$$

The distinction between these scalar structures and the Frobenius-module structures is essential in using the preceding cohomology calculation. Corollary 5.4 supplies the Laurent cocycle vanishings and finite-dimensionality of global one-forms that we use to make $\text{Pic}(|D|, \mathcal{R})$ p -divisible. Orders along the components of D then let intersection with H descend to a nonzero integer degree on that same group. These two properties are incompatible.

6.1 Global Cartier adjustment

Lemma 6.1. *Every global Laurent one-form is closed. On the finite-dimensional k -vector space*

$$W = H^0(|D|, \Omega_{\mathcal{R}}^1),$$

Cartier is a bijective inverse-Frobenius-semilinear map, and $C - \text{id} : W \rightarrow W$ is surjective as an additive map. Moreover, every additive $\mathcal{Z}^1$ -valued Čech one-cocycle is a coboundary after refining its open cover.

Proof. Corollary 5.4 gives finite-dimensionality of W and actual Laurent one-cocycle vanishing for $\mathcal{B}^1$ and $\mathcal{Z}^1$, proving the last assertion. To lift a global form through C , choose local lifts in $\mathcal{Z}^1$ using Equation (6.1). Their differences form a $\mathcal{B}^1$ -cocycle. Subtracting a coboundary makes the lifts agree, so

$$C : H^0(|D|, \mathcal{Z}^1) \longrightarrow W$$

is surjective. Its source is a vector subspace of W . Since the ground field is perfect, a semilinear map has the same dimension properties as a linear map. Finite-dimensionality therefore implies $H^0(\mathcal{Z}^1) = W$ and bijectivity of C .

For the last assertion about C , set $F = C^{-1}$, a bijective p -semilinear map. Choose coordinates on $W \simeq k^n$. Then $F(x) = Ax^{[p]}$ with $A \in \text{GL}_n(k)$, where $x^{[p]}$ means componentwise p -th powers. The polynomial map $F - \text{id}$ extends to

$$\Psi : \mathbb{P}^n \longrightarrow \mathbb{P}^n, \quad [x : t] \longmapsto [Ax^{[p]} - t^{p-1}x : t^p]. \quad (6.2)$$

There is no base point: if $t = 0$, the first coordinates vanish only when $x = 0$. Also $\Psi^*\mathcal{O}_{\mathbb{P}^n}(1) = \mathcal{O}_{\mathbb{P}^n}(p)$. A positive-dimensional fiber would contain a projective curve on which this line has both degree zero and positive degree, a contradiction. Hence Ψ is finite. Its image is closed and has dimension n , so it is all of $\mathbb{P}^n$. The inverse image of the standard affine chart is exactly that chart, proving surjectivity of $F - \text{id}$ over the algebraically closed field. The case $n = 0$ is immediate. Applying F to $(C - \text{id})\beta = \eta$ gives the equivalent equation

$$(F - \text{id})\beta = -F\eta,$$

so $C - \text{id}$ is surjective as well. $\square$

Lemma 6.2. *Every $\mathcal{R}$ -line bundle admits an integrable connection whose local connection one-forms are fixed by Cartier.*

Proof. Let $\mathcal{L}$ be such a line bundle. Its frame cover can be refined by opens $U_i \cap |D|$, where $U_i \subset V$ is affine, has étale coordinates, makes D principal, and trivializes the finitely many algebraic bundles in use. Quasi-compactness of $|D|$ permits a finite such cover. Its finite intersections are restrictions of affine opens, since V is separated. On each intersection, a Laurent coefficient has a finite pole bound: take a finite subcover of local bounds and glue in their common completed lattice, as in Section 4. Taking the maximum over the finitely many coefficients gives a common upper cut for a cochain. These observations also apply after finite refinements for local Cartier lifts. Thus the cocycles below lie in the actual Laurent complexes to which the vanishing results apply.

Choose frames e_i on this cover, and write $e_j = g_{ij}e_i$ on overlaps. Then $g_{ij}g_{jk} = g_{ik}$, so

$$d \log g_{ij} = g_{ij}^{-1} dg_{ij}$$

is a closed additive one-cocycle. It is Cartier-fixed. Indeed, the intrinsic inverse Cartier formula gives

$$C^{-1}(g^{-1}dg) = [g^{-p}g^{p-1}dg] = [g^{-1}dg]$$

for every Laurent unit g . By Lemma 6.1, after refining the cover there are closed one-forms α_i such that

$$\alpha_j - \alpha_i = d \log g_{ij}.$$

These are the connection forms of a connection defined by $\nabla e_i = \alpha_i \otimes e_i$. Their closedness is precisely its integrability, since the bundle has rank one.

The forms $C\alpha_i - \alpha_i$ agree on overlaps and hence define a global form $\eta \in W$. Choose $\beta \in W$ with $(C - \text{id})\beta = \eta$, using Lemma 6.1. Replace every α_i by $\alpha_i - \beta$. This preserves their differences and their closedness, and now $C\alpha_i = \alpha_i$ for all i . $\square$

6.2 Horizontal sections over a Laurent chart

We give the local descent calculation, including its compatibility with restriction. This is the algebra underlying Cartier descent [19, Theorem 5.1 and formulas (5.1.2)–(5.1.7)]; the formulas below apply directly to our Laurent rings by their retained p -basis, without a finite-type hypothesis on those rings.

Take a smooth affine chart $U = \text{Spec } A \subset V$ with étale coordinates x_1, x_2, x_3 , and let f be an equation of D on it. Put

$$B = \varprojlim_n A/(f^n), \quad R = B[1/f], \quad S = R^p.$$

Lemma 5.3 gives

$$R = \bigoplus_{0 \leq I_1, I_2, I_3 < p} Sx^I, \quad \Omega_{R/k}^1 = \bigoplus_{i=1}^3 R dx_i. \quad (6.3)$$

Write ∂_i for the coordinate derivations. They commute, kill S , and satisfy $\partial_i^p = 0$.

Lemma 6.3 (The rank-one curvature calculation). *If $\alpha = \sum_i a_i dx_i$ is closed and $C\alpha = \alpha$, then the connection operators*

$$D_i = \partial_i + a_i : R \longrightarrow R$$

commute and satisfy

$$D_i^p = 0, \quad [D_i, x_j] = \delta_{ij}. \quad (6.4)$$

Here a ring element in an operator formula denotes multiplication by that element.

Proof. If $C\alpha = \sum_i b_i dx_i$, the p -basis Cartier calculation represents α in the form

$$\alpha = dc + \sum_i b_i^p x_i^{p-1} dx_i \quad (c \in R).$$

Consequently

$$\partial_i^{p-1} a_i = (p-1)! b_i^p = -b_i^p. \quad (6.5)$$

Exact terms disappear because $\partial_i^p c = 0$, and b_i^p is constant for every coordinate derivation. The Cartier-fixed condition therefore says $a_i^p + \partial_i^{p-1} a_i = 0$.

We recall the operator identity giving its significance. For a derivation ∂ with $\partial^p = 0$ and an element $a \in R$,

$$(\partial + a)^p = a^p + \partial^{p-1} a. \quad (6.6)$$

First, the left side commutes with multiplication by any $b \in R$: in characteristic p , its commutator is the p -fold iterated commutator, namely multiplication by $\partial^p b = 0$. It is therefore multiplication by its value on 1. Over the integers, successive use of the product rule expands $(\partial + a)^p(1)$ as a sum indexed by set partitions of $\{1, \dots, p\}$, a block of size j contributing $\partial^{j-1} a$. This formula follows by adjoining the last label either as a singleton or to

one existing block. A cyclic permutation of the p labels has orbits of size p on these partitions, except for the discrete partition and the partition with one block. To check the exceptions, in an invariant partition a block is either the whole set or has an orbit of p members, in which case all blocks are singletons. The summands have the same value throughout an orbit, so only a^p and $\partial^{p-1}a$ survive in characteristic p . This proves Equation (6.6) and hence $D_i^p = 0$. Finally,

$$[D_i, D_j] = \partial_i a_j - \partial_j a_i = 0$$

by closedness, and the commutator with x_j follows directly from the Leibniz rule. $\square$

Lemma 6.4 (Horizontal expansion). *Let M be a free rank-one R -module with a connection whose operators satisfy Equation (6.4) and commute. Its horizontal module*

$$H_M = \bigcap_i \ker(D_i : M \rightarrow M)$$

is a finite projective rank-one S -module, and multiplication gives an isomorphism

$$R \otimes_S H_M \xrightarrow{\sim} M. \quad (6.7)$$

Formation of H_M commutes with restriction to a smaller Laurent chart retaining the coordinates.

Proof. Define S -linear operators

$$P_i = \sum_{n=0}^{p-1} \frac{(-1)^n}{n!} x_i^n D_i^n, \quad P = P_1 P_2 P_3. \quad (6.8)$$

The multiplication operator x_i^n is placed on the left. All factorials here are invertible in k . The commutator relation gives a telescoping sum

$$D_i P_i = \frac{(-1)^{p-1}}{(p-1)!} x_i^{p-1} D_i^p = 0.$$

Also P_i is the identity on $\ker D_i$, and distinct P_i 's commute with each other and with D_j for $j \neq i$. Hence P is an idempotent with image H_M .

For one coordinate, the binomial identity gives

$$\sum_{n=0}^{p-1} \frac{x_i^n}{n!} P_i D_i^n = \text{id}_M. \quad (6.9)$$

Indeed, after substituting Equation (6.8), the coefficient of $x_i^r D_i^r$, for $0 \leq r < p$, is

$$\sum_{n+j=r} \frac{(-1)^j}{n!j!} = \begin{cases} 1 & r = 0, \\ 0 & r > 0. \end{cases}$$

All terms with $r \geq p$ vanish by $D_i^p = 0$. Multiplying Equation (6.9) over the three coordinates gives the explicit expansion

$$m = \sum_{0 \leq I_j < p} x^I h_I, \quad h_I = \frac{1}{I!} P(D^I m) \in H_M. \quad (6.10)$$

Here $I! = I_1! I_2! I_3!$ and $D^I = D_1^{I_1} D_2^{I_2} D_3^{I_3}$. The coefficients are unique: for $h \in H_M$,

$$P D^J (x^I h) = \begin{cases} I! h & I = J, \\ 0 & I \neq J, \end{cases} \quad 0 \leq I_i, J_i < p.$$

This follows by differentiation and the same binomial sum, which makes $P(x^K h) = 0$ for every nonzero K in that range. Together with Equation (6.3), this proves Equation (6.7).

As an S -module, M is free of rank p^3 . The image of the idempotent P is therefore finite projective. The extension $S \subset R$ is finite free and faithfully flat; Equation (6.7) shows that this projective module has rank one.

For restriction, let $R \rightarrow R'$ be the homomorphism of Laurent chart rings and set $S' = (R')^p$. The shared monomial bases give an isomorphism

$$S' \otimes_S R \xrightarrow{\sim} R'.$$

The restricted module is consequently $S' \otimes_S M$, its operators are $1 \otimes D_i$, and its projector is $1 \otimes P$. Taking the image of a split idempotent commutes with every scalar extension, so its horizontal module is $S' \otimes_S H_M$. This argument does not require a separate flatness assertion for the restriction homomorphism. Since horizontality means $\nabla m = 0$, the construction is also independent of the coordinate system and the chosen line frame. $\square$

6.3 The formal Zariski topology and Picard divisibility

The projective module in Lemma 6.4 must be locally trivial on the topology of $|D|$. The next elementary completion fact supplies this step.

Lemma 6.5. *For the smooth affine chart and rings A, B, R above, B is a regular Noetherian ring. Every finite projective rank-one R -module extends to a finite projective rank-one B -module. Its associated line is locally trivial after restriction to formal Zariski neighborhoods of every point of $U \cap |D|$. The same assertions hold for $B^p \subset R^p$.*

Proof. The Noetherian completion theorem gives that B is Noetherian and flat over A , and $B/(f^n) = A/(f^n)$ [28, Lemmas 10.97.1–10.97.5]. The element f belongs to the Jacobson radical of B : an element congruent to 1 modulo f has an inverse by the convergent geometric series. More generally, an element invertible modulo f is invertible in B [28, Lemma 10.96.6].

Every maximal ideal $\mathfrak{n} \subset B$ thus corresponds to a maximal ideal $\mathfrak{m} \subset A$ containing f , and $\mathfrak{n} = \mathfrak{m}B$. For each s , completion modulo $\mathfrak{m}^s$ does nothing, since f^s already vanishes there. Hence

$$B/\mathfrak{n}^s \simeq A/\mathfrak{m}^s, \quad \widehat{B}_{\mathfrak{n}} \simeq \widehat{A}_{\mathfrak{m}}.$$

The latter ring is regular, as $A_{\mathfrak{m}}$ is a smooth local ring. A Noetherian local ring is regular if and only if its maximal-ideal completion is regular [28, Lemma 15.44.4], so every $B_{\mathfrak{n}}$ is regular. Localization now gives regularity at every prime of B .

A regular Noetherian ring is locally factorial [28, Lemma 15.123.2]. Its finitely many irreducible components are disjoint, since each local ring is a domain. On each component choose a rational section of the given line on $\text{Spec } R$ and take its Weil divisor. The closure of this finite Weil divisor on $\text{Spec } B$ is Cartier by local factoriality. Its line bundle restricts to the original one. Doing this on all components gives the asserted extension. This is the Cartier–Weil correspondence on a regular scheme [15, II, §6]; global unique factorization of B is not required.

Let $z \in U \cap |D|$. The extended line is free on a basic open $\text{Spec } B_g$ containing the prime corresponding to z . Choose $g_0 \in A$ whose image modulo f equals that of g . Then $g_0(z) \neq 0$. On the formal basic neighborhood coming from $\text{Spec } A_{g_0}$, its ring is $B' = \widehat{A}_{g_0}$, and the image of g is invertible: modulo f it equals the invertible element g_0 . The line is therefore free on this formal neighborhood, and remains free after inverting f . Finally, Frobenius is an abstract ring isomorphism from B onto B^p , and from R onto R^p , because these rings are reduced. It identifies the corresponding formal basic neighborhoods. Thus the same proof applies to the power subrings. $\square$

Proposition 6.6 (Divisibility of the Laurent Picard group). *Multiplication by p on $\text{Pic}(|D|, \mathcal{R})$ is surjective.*

Proof. Give an arbitrary $\mathcal{R}$ -line bundle $\mathcal{L}$ the connection from Lemma 6.2. On a chart with a line frame, Lemma 6.3 gives commuting nilpotent connection operators, and Lemma 6.4 makes the horizontal module finite projective of rank one over R^p . Its restriction compatibility and intrinsic definition glue these modules into a sheaf $\mathcal{H} = \ker \nabla$ over $\mathcal{R}^p$. Lemma 6.5 shows that $\mathcal{H}$ is locally free of rank one in the required formal Zariski topology. The local multiplication isomorphisms glue to

$$\mathcal{R} \otimes_{\mathcal{R}^p} \mathcal{H} \simeq \mathcal{L}.$$

Transport $\mathcal{H}$ through the sheaf isomorphism $\mathcal{R} \rightarrow \mathcal{R}^p$, $a \mapsto a^p$, to obtain a line bundle $\mathcal{N}$ over $\mathcal{R}$. Extension from $\mathcal{R}^p$ to $\mathcal{R}$ is its absolute Frobenius pullback. In local frames, that pullback raises the transition units of $\mathcal{N}$ to their p -th powers, which are also the transition units of $\mathcal{N}^{\otimes p}$. Thus

$$\mathcal{L} \simeq \mathcal{N}^{\otimes p}.$$

This is a global root on the stated ringed space, proving the surjectivity assertion. $\square$

6.4 Orders of units and an integral degree

Infinite Frobenius divisibility forces integral numerical invariants to vanish; this is also the numerical mechanism for stratified bundles in [8, Lemma 2.1 and Corollary 2.2]. Here the required degree is defined first on the completion. We must prove that it descends to the Laurent Picard group before applying that elementary divisibility obstruction.

Lemma 6.7. *Let $\iota_\alpha : D_\alpha \hookrightarrow |D|$ be the inclusion of a prime component. There is an exact sequence of abelian sheaves*

$$1 \longrightarrow \widehat{\mathcal{O}}^\times \longrightarrow \mathcal{R}^\times \xrightarrow{\text{ord}} \bigoplus_{\alpha} (\iota_\alpha)_* \mathbb{Z} \longrightarrow 0. \quad (6.11)$$

Consequently restriction gives a surjection

$$\text{Pic}(|D|, \widehat{\mathcal{O}}) \longrightarrow \text{Pic}(|D|, \mathcal{R}),$$

whose kernel is generated by the completed restrictions of $\mathcal{O}_V(D_\alpha)$.

Proof. Work near a point on an affine chart as above, small enough that each component through it has a prime equation f_α , and write f as a unit times $\prod_\alpha f_\alpha^{m_\alpha}$. Every f_α remains a non-zero-divisor in B , by flatness of completion. It also remains prime, because

$$B/(f_\alpha) \simeq A/(f_\alpha). \quad (6.12)$$

Indeed, the induced f -adic topology on $A/(f_\alpha)$ is the zero-ideal topology, so completion leaves this integral ring unchanged.

If $u \in B[1/f]^\times$, write $u = a/f^r$ and $u^{-1} = b/f^s$, with $a, b \in B$. Since f is a non-zero-divisor, $ab = f^{r+s}$ in B . Successively use primality to cancel each of the finitely many prime factors of the right side from either a or b . At the end the remaining factors multiply to a unit. This expresses u uniquely as

$$u = v \prod_{\alpha} f_{\alpha}^{n_{\alpha}}, \quad v \in B^\times, \quad n_{\alpha} \in \mathbb{Z}. \quad (6.13)$$

Uniqueness follows from the same prime cancellation. The exponents agree on smaller charts and are locally constant along the respective D_α . They define the order map. Its kernel consists exactly of formal units, and the local equations give its local surjectivity, proving Equation (6.11).

On an irreducible space with its Zariski topology, the constant sheaf $\mathbb{Z}$ is flasque: every nonempty open is irreducible, its locally constant integer-valued functions are constant, and all restriction maps are surjective. Each pushed-forward sheaf $(\iota_\alpha)_*\mathbb{Z}$, and their finite direct sum, is therefore flasque. Its first cohomology vanishes. Line bundles on a ringed space are classified by the first cohomology of its unit sheaf. The cohomology sequence of Equation (6.11) consequently proves surjectivity of the Picard restriction map. The image of the connecting map from the global integer sections is the subgroup generated by $\mathcal{O}_V(D_\alpha)$, as is seen from their local equations. Exactness proves the kernel assertion. $\square$

Proposition 6.8. *The prepared cycle of Proposition 2.1 cannot satisfy the conclusions of Corollary 5.4.*

Proof. For a line bundle $\mathcal{L}$ on the completion, its restriction to each integral projective surface D_α is an ordinary line bundle. Define the integer

$$\Phi(\mathcal{L}) = \sum_{\alpha} m_{\alpha} (c_1(\mathcal{L}|_{D_{\alpha}}) \cdot H|_{D_{\alpha}}). \quad (6.14)$$

Intersection with the Cartier class H gives an integral degree on each surface, so Φ is an additive homomorphism to $\mathbb{Z}$. The numerical properties of the prepared cycle give

$$\Phi(\mathcal{O}_V(D_{\beta})) = \sum_{\alpha} m_{\alpha} D_{\alpha} D_{\beta} H = D D_{\beta} H = 0.$$

Lemma 6.7 therefore makes Φ descend to a homomorphism

$$\bar{\Phi} : \text{Pic}(|D|, \mathcal{R}) \longrightarrow \mathbb{Z}.$$

It is nonzero, since

$$\bar{\Phi}(\mathcal{O}_V(H)|_{\mathcal{R}}) = D H^2 > 0. \quad (6.15)$$

On the other hand, Proposition 6.6 implies that every class in this Picard group is divisible by p^n for every positive integer n . Its image under $\bar{\Phi}$ must therefore be an integer divisible by every p^n , hence zero. This contradicts Equation (6.15). $\square$

Completion of the proof of Theorem 1.1. The assumption of maximal nef dimension supplied the prepared cycle and all the conclusions of Corollary 5.4. Proposition 6.8 rules out that assumption. Thus $n(X, K_X) \leq 2$ over the uncountable algebraically closed extension used in the preparation. The nef-dimension-at-most-two abundance theorem [32, Theorem 1.8] applies to the projective log canonical threefold pair $(X, 0)$ in characteristic $p > 3$ with nef canonical divisor. It makes K_X semiample there.

To finish over the original field, denote it by k_0 and the extension by k . Restore X for the original threefold and write $X_k = X \times_{k_0} k$. Choose a globally generated Cartier multiple of K_{X_k} , then take a further positive multiple divisible by the Cartier index of K_X over k_0 . Call the resulting integer m . The line bundle $\mathcal{O}_{X_k}(mK_{X_k})$ is still globally generated and is not numerically trivial, since $\nu(K_{X_k}) = 1$. It has at least two independent sections: if a globally generated line bundle has only one, that section trivializes it, contradicting its nonzero numerical class.

Compatibility of the canonical Cartier multiple with field extension and proper section-space base change give

$$H^0(X, \mathcal{O}_X(mK_X)) \otimes_{k_0} k \simeq H^0(X_k, \mathcal{O}_{X_k}(mK_{X_k})).$$

Their dimensions are equal, proving the required inequality over k_0 . In particular $\kappa(X, K_X) \geq 1$; nefness and $\nu(K_X) = 1$ give $\kappa(X, K_X) = 1$, and the established positive-Kodaira-dimension abundance theorem [32, Theorem 1.6] also gives semiampleness over the original field. $\square$

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